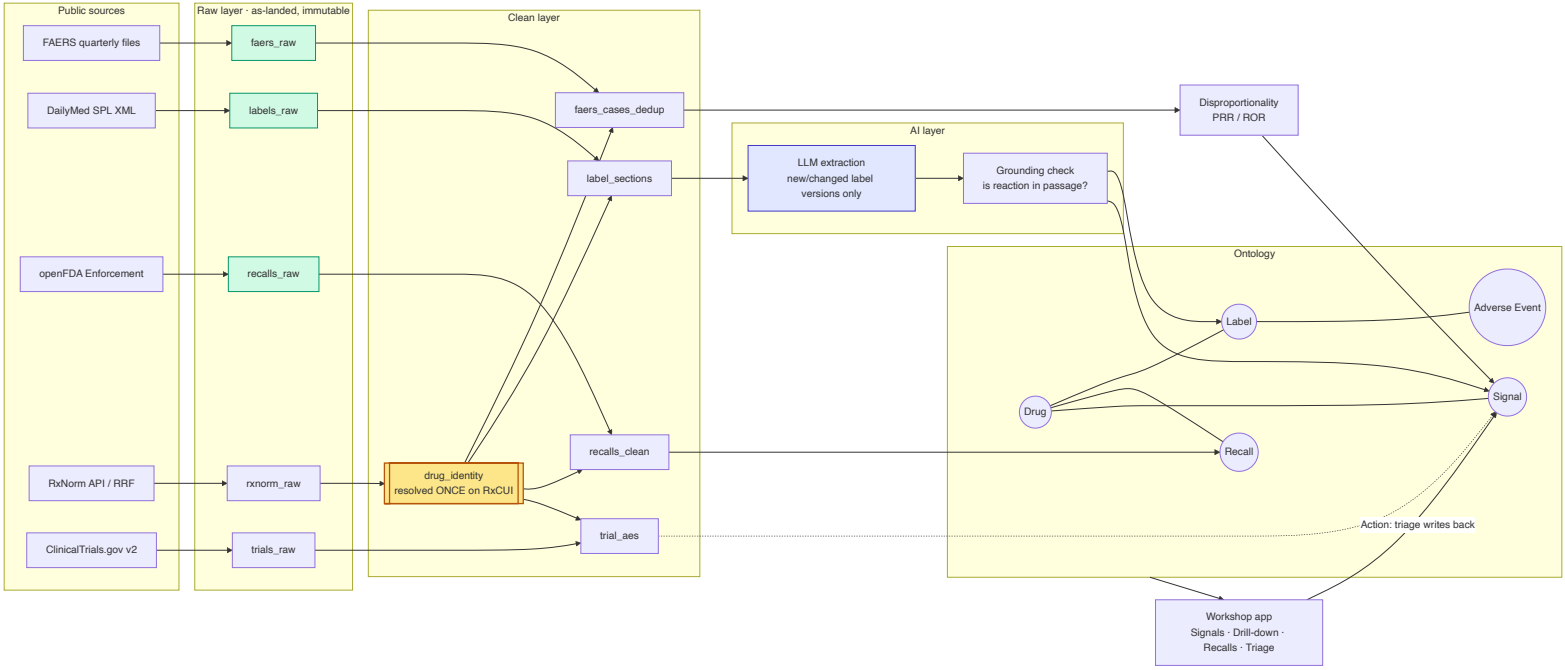
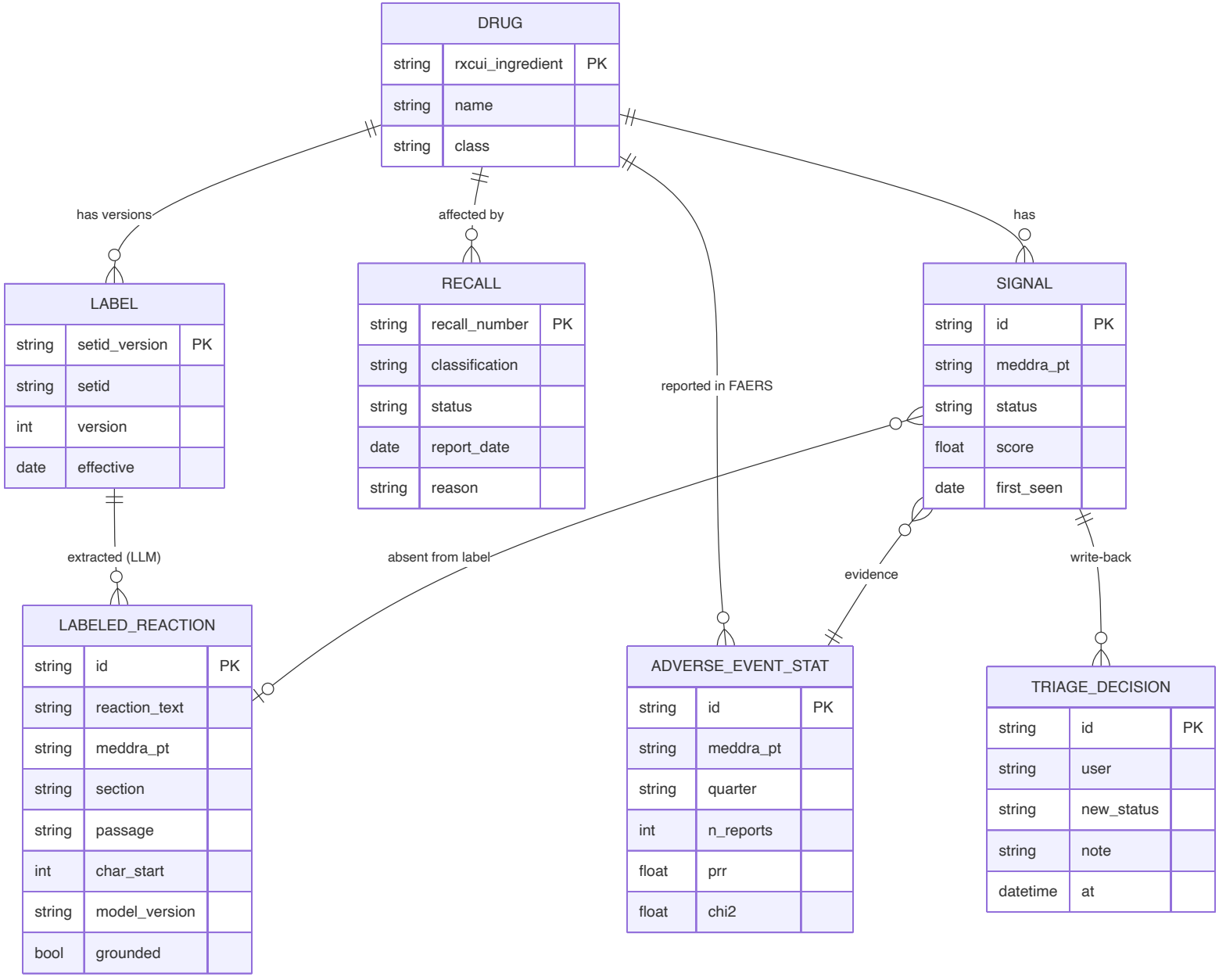
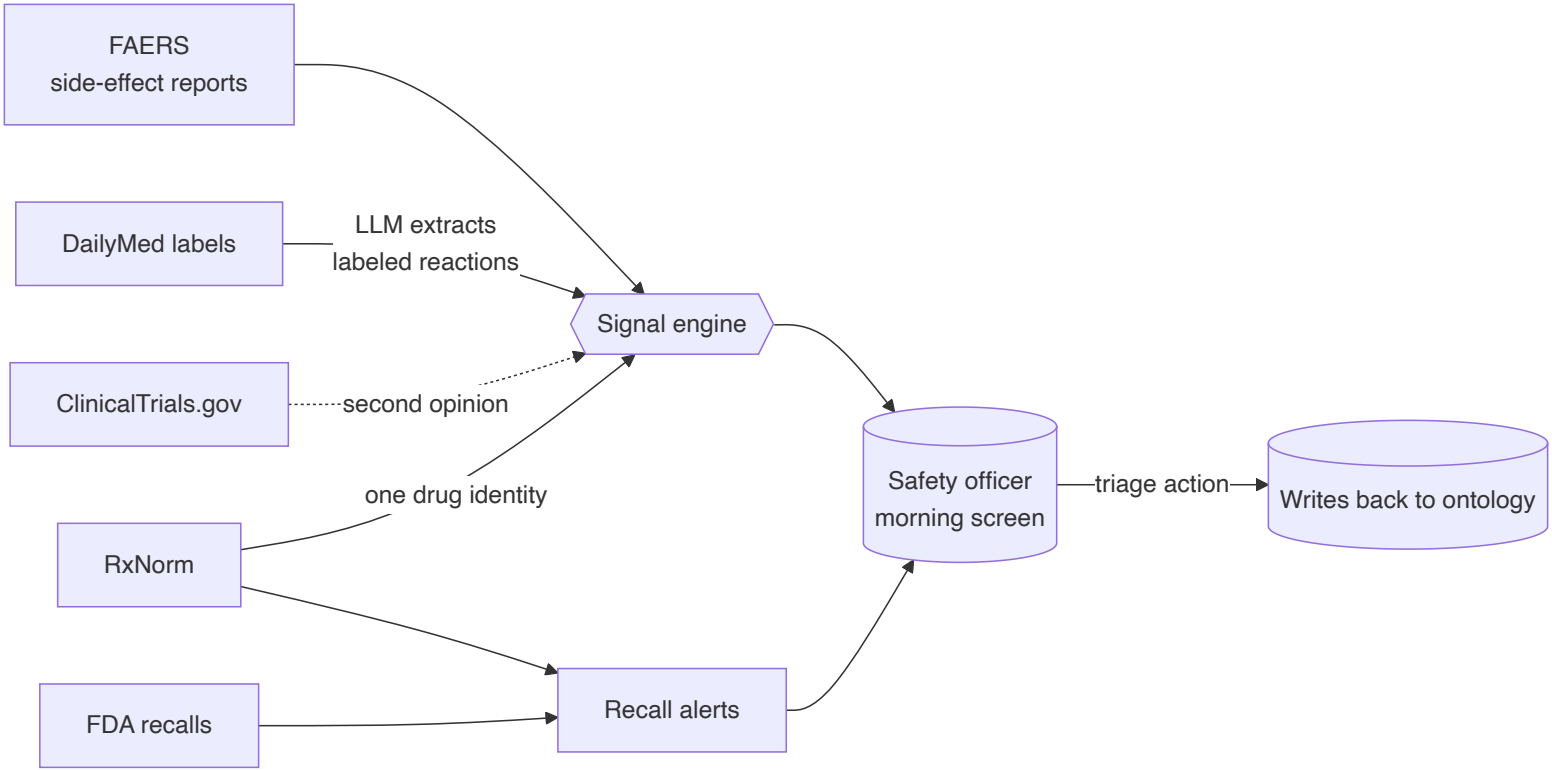
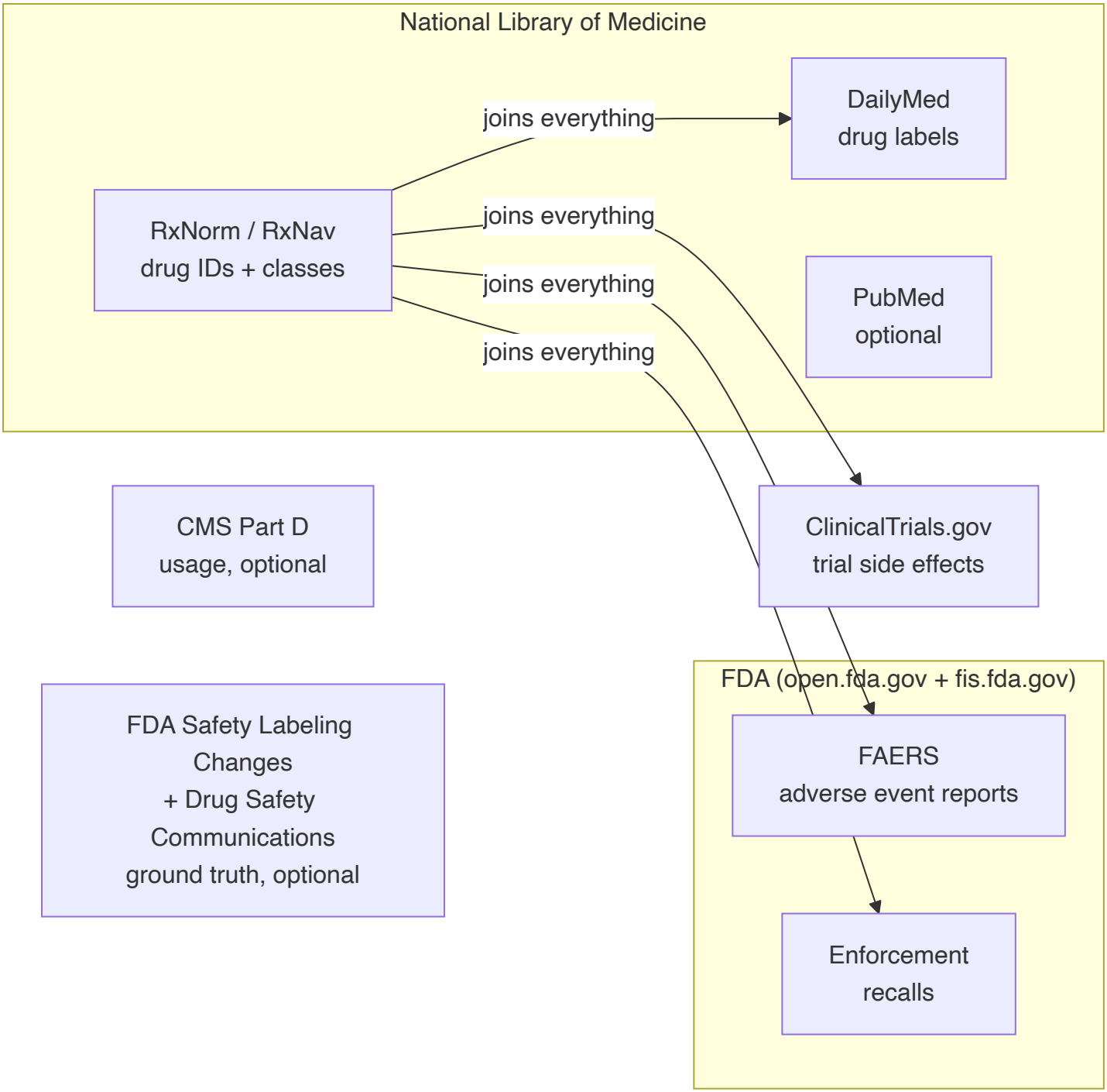
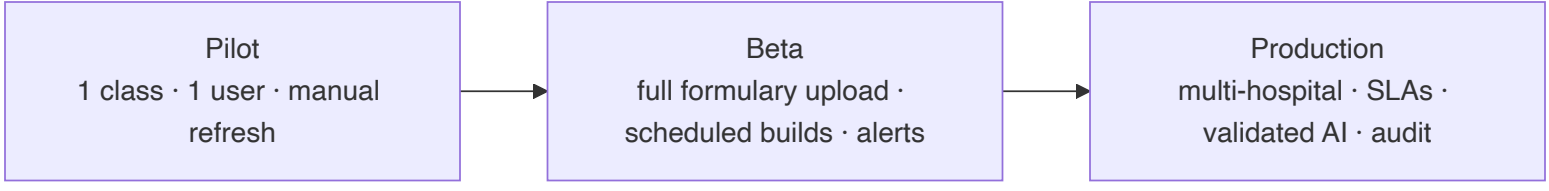

0 · Discover
Empathize
1 · Define
Scope + source profiling
2 · Pilot build
Pipelines → Ontology → AI
→ App
3 · Harden + Prove
Eval, optimization log,
monitoring
4 · Demo + Note
20-min demo, 1-page
architecture
5 · Scale path
Production roadmap


Public sources
FAERS quarterly files
DailyMed SPL XML
openFDA Enforcement
RxNorm API / RRF
ClinicalTrials.gov v2
Raw layer · as-landed, immutable
faers_raw
labels_raw
recalls_raw
rxnorm_raw
trials_raw
Clean layer
faers_cases_dedup
label_sections
recalls_clean
drug_identity
resolved ONCE on RxCUI
trial_aes
AI layer
LLM extraction
new/changed label
versions only
Grounding check
is reaction in passage?
Disproportionality
PRR / ROR
Ontology
Label
Adverse Event
Drug
Signal
Recall
Action: triage writes back
Workshop app
Signals · Drill-down ·
Recalls · Triage

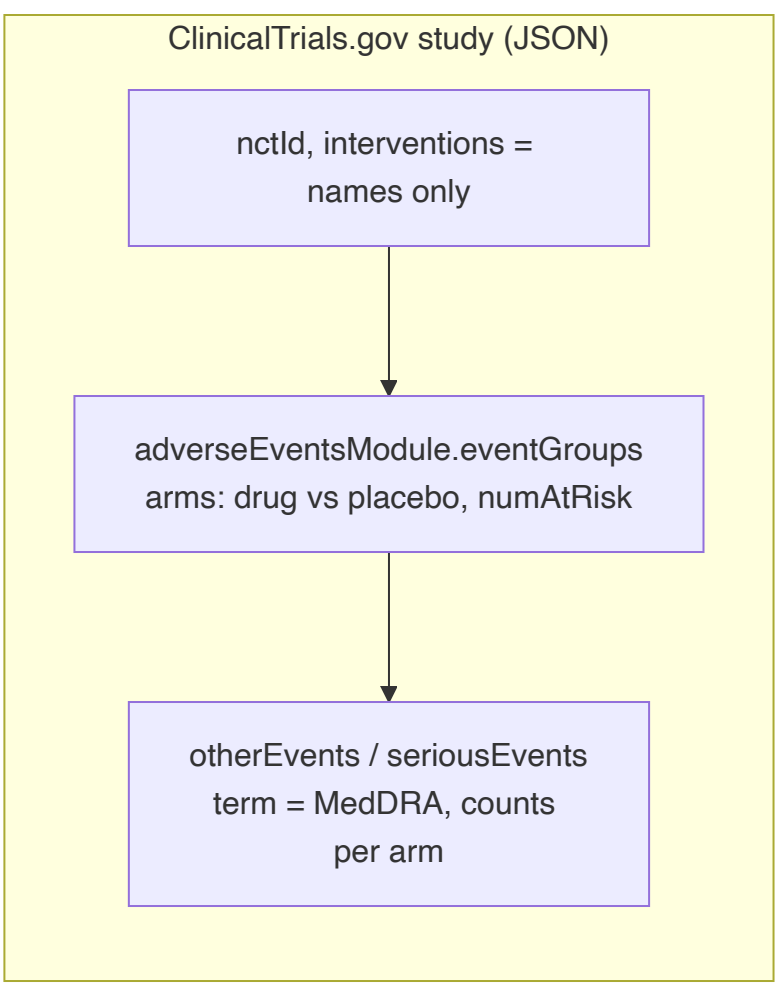

DRUG
string rxcui_ingredient PK
string name
string class
has versions
affected by
has
reported in FAERS
LABEL
string setid_version PK
string setid
int version
date effective
RECALL
string recall_number PK
string classification
string status
date report_date
string reason
SIGNAL
string id PK
string meddra_pt
string status
float score
date first_seen
extracted (LLM)
absent from label
evidence
write-back
LABELED_REACTION
string id PK
string reaction_text
string meddra_pt
string section
string passage
int char_start
string model_version
bool grounded
ADVERSE_EVENT_STAT
string id PK
string meddra_pt
string quarter
int n_reports
float prr
float chi2
TRIAGE_DECISION
string id PK
string user
string new_status
string note
datetime at

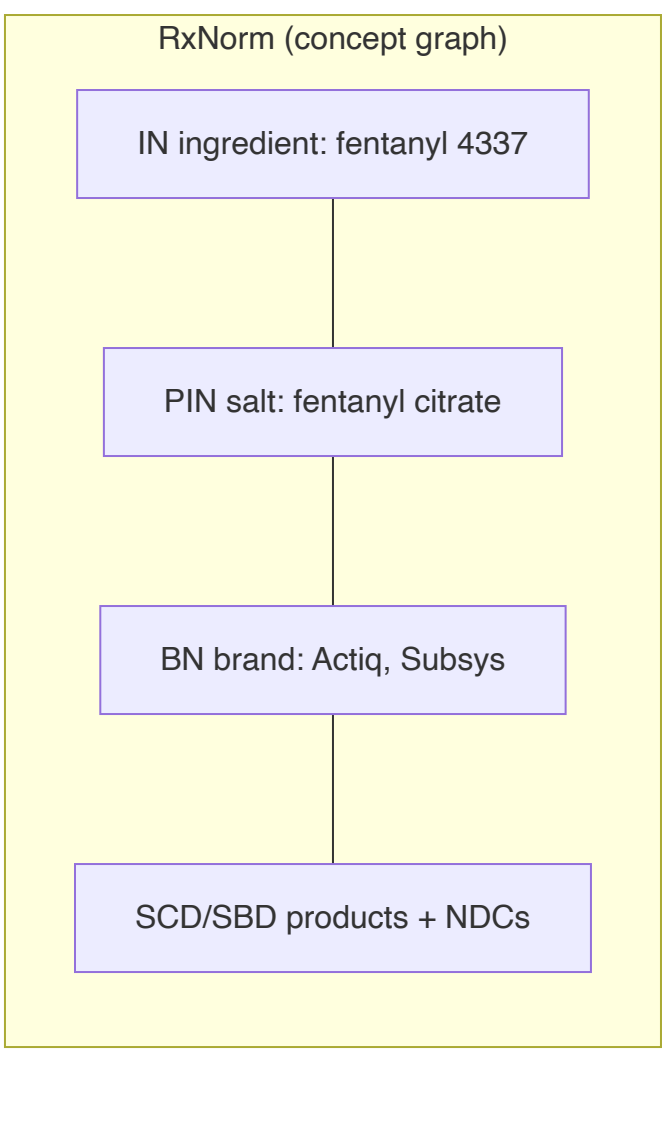

FAERS
side-effect reports
DailyMed labels
LLM extracts
labeled reactions
ClinicalTrials.gov
second opinion
one drug identity
RxNorm
FDA recalls
Signal engine
Recall alerts
Safety officer
morning screen
triage action
Writes back to ontology

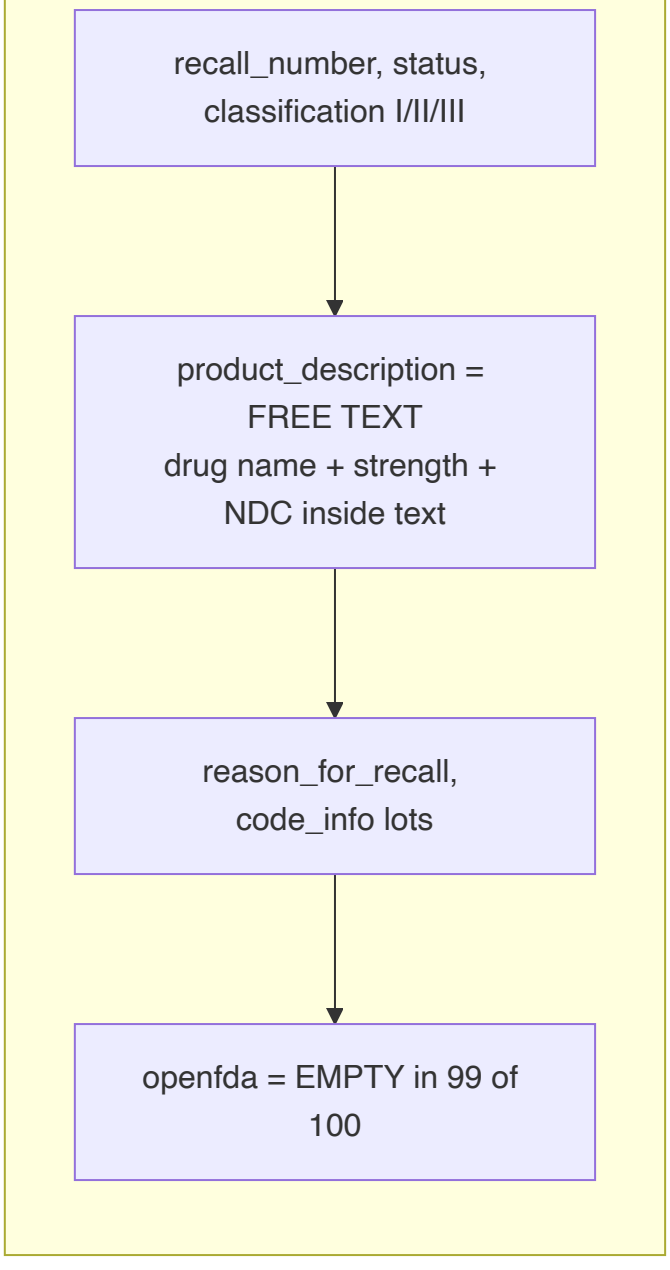

National Library of Medicine
DailyMed
drug labels
joins everything
RxNorm / RxNav
drug IDs + classes
joins everything
joins everything
joins everything
PubMed
optional
CMS Part D
usage, optional
ClinicalTrials.gov
trial side effects
FDA (open.fda.gov + fis.fda.gov)
FAERS
adverse event reports
FDA Safety Labeling
Changes
+ Drug Safety
Communications
ground truth, optional
Enforcement
recalls

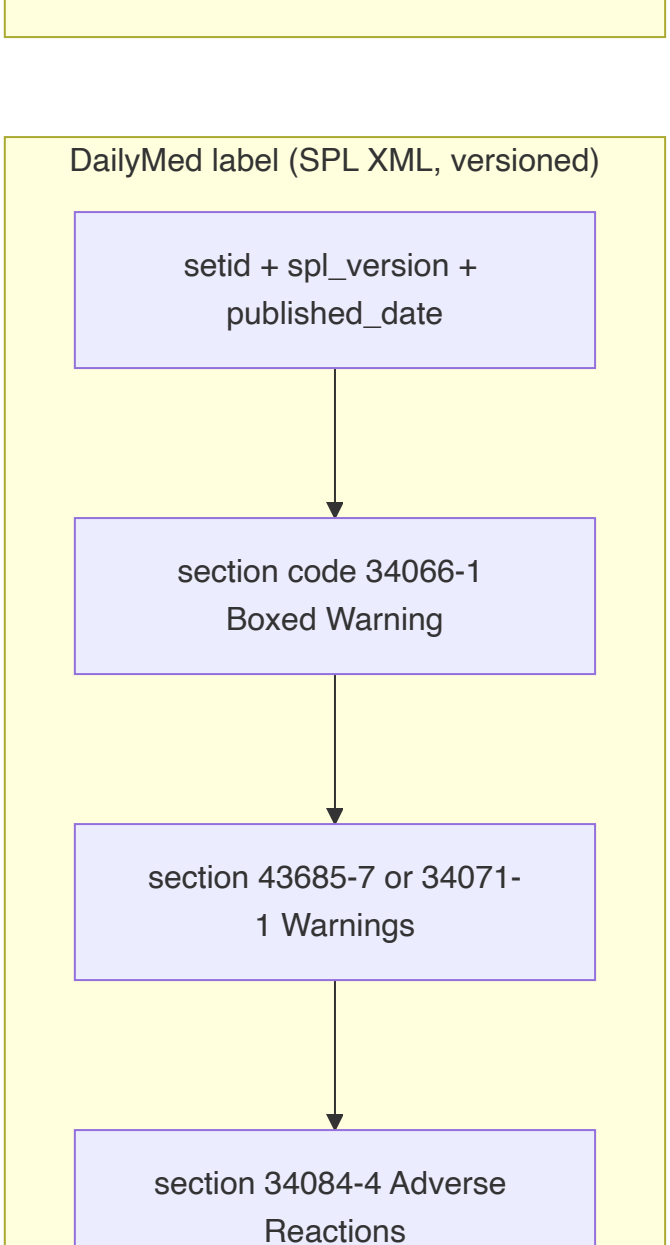

Pilot
1 class · 1 user · manual refresh
Beta
full formulary upload · scheduled builds · alerts
Production
multi-hospital · SLAs · validated AI · audit

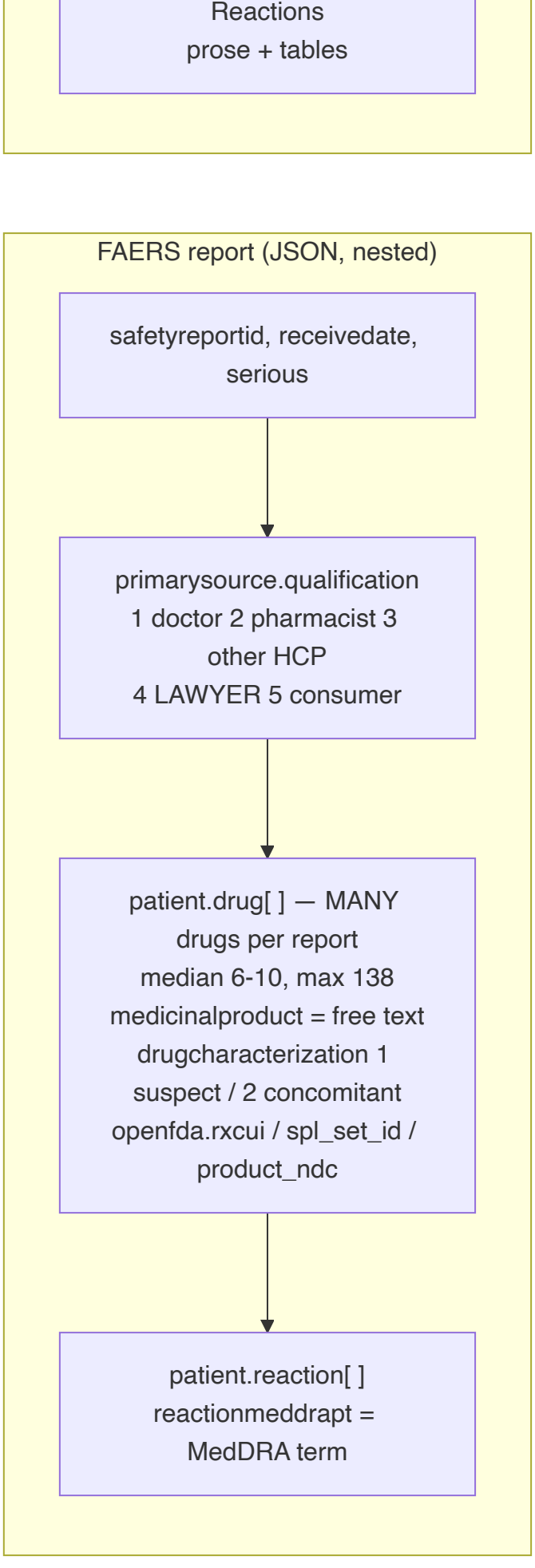
How each source is shaped (one record = one box)



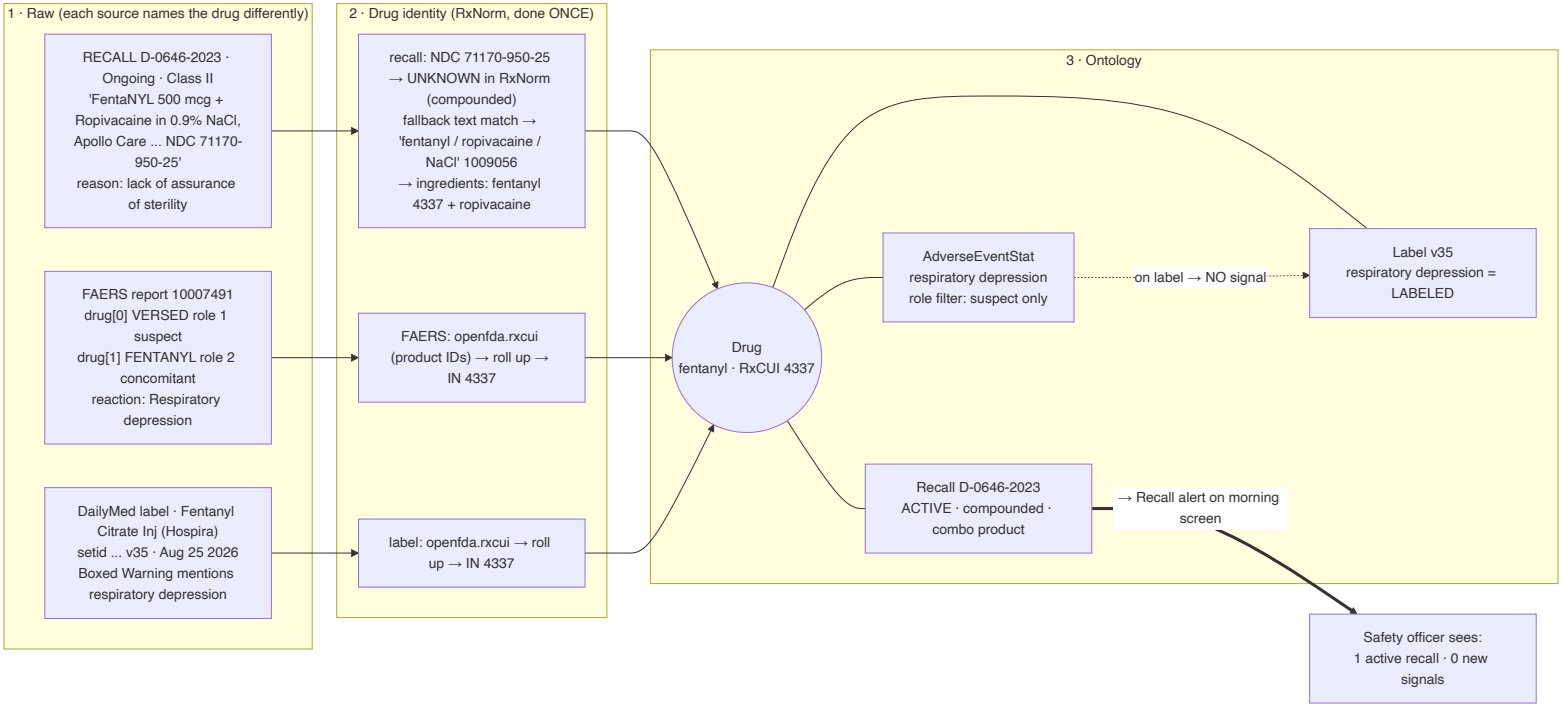




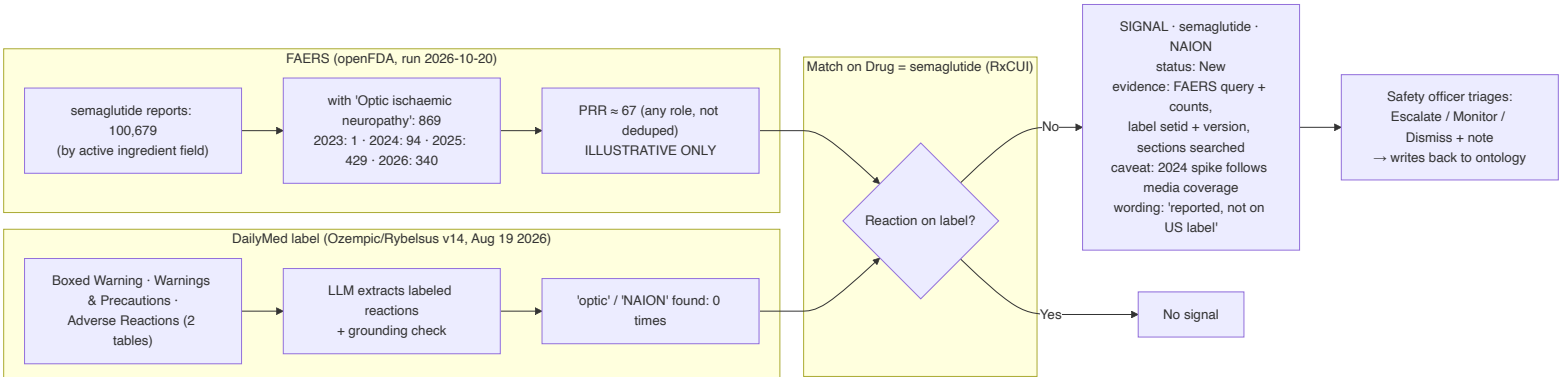


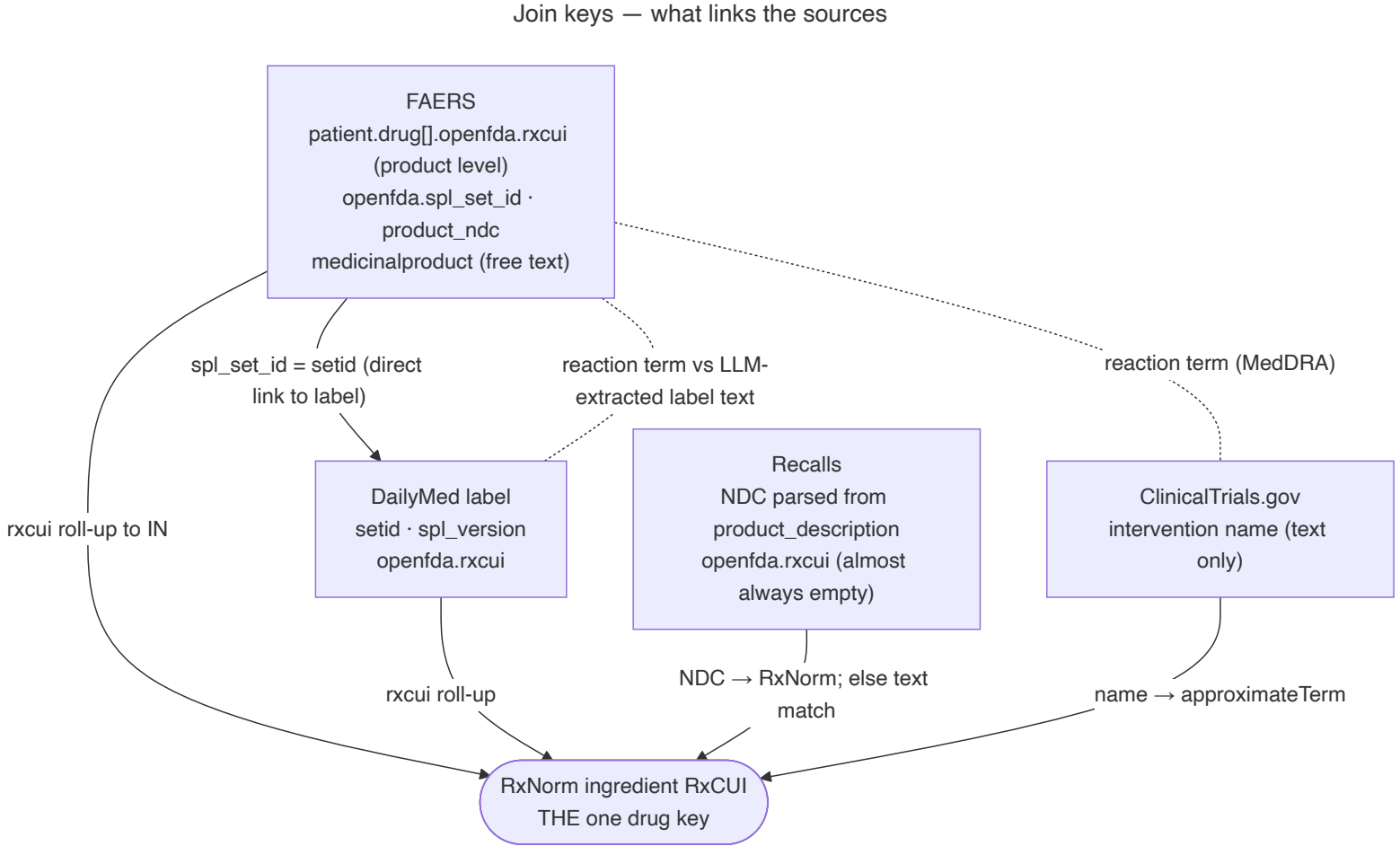


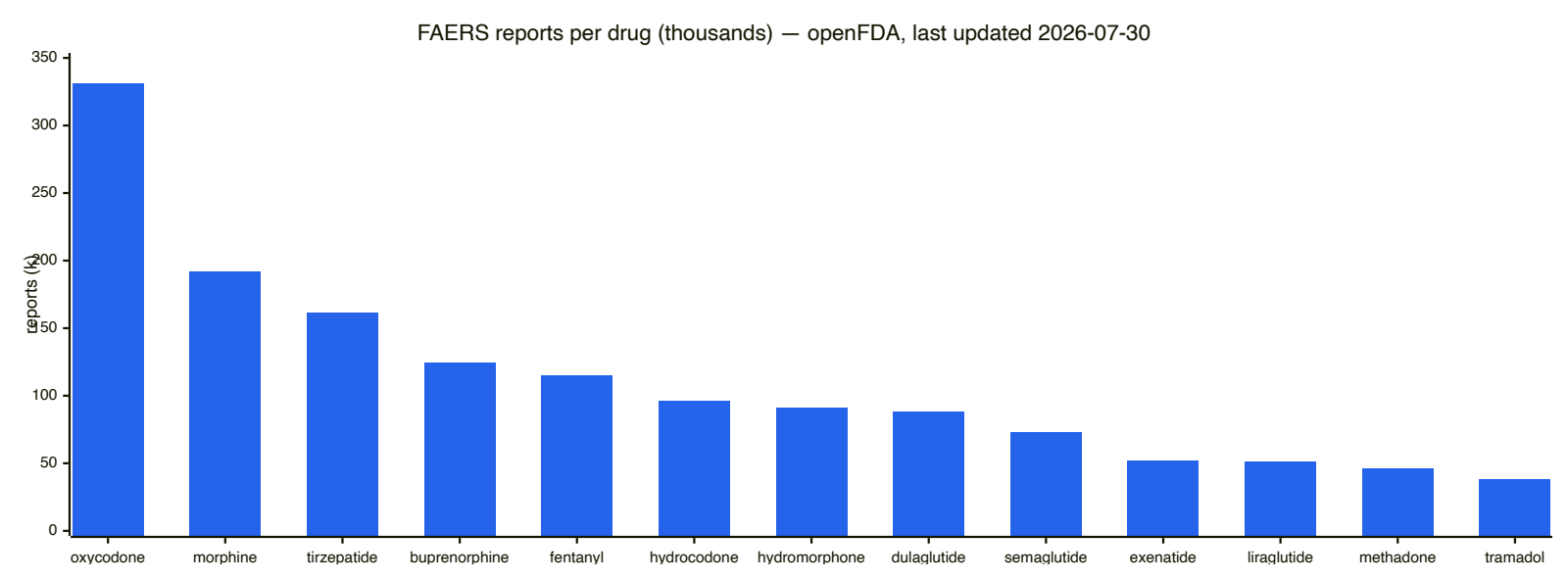
Example A — fentanyl: three sources, three shapes, one Drug

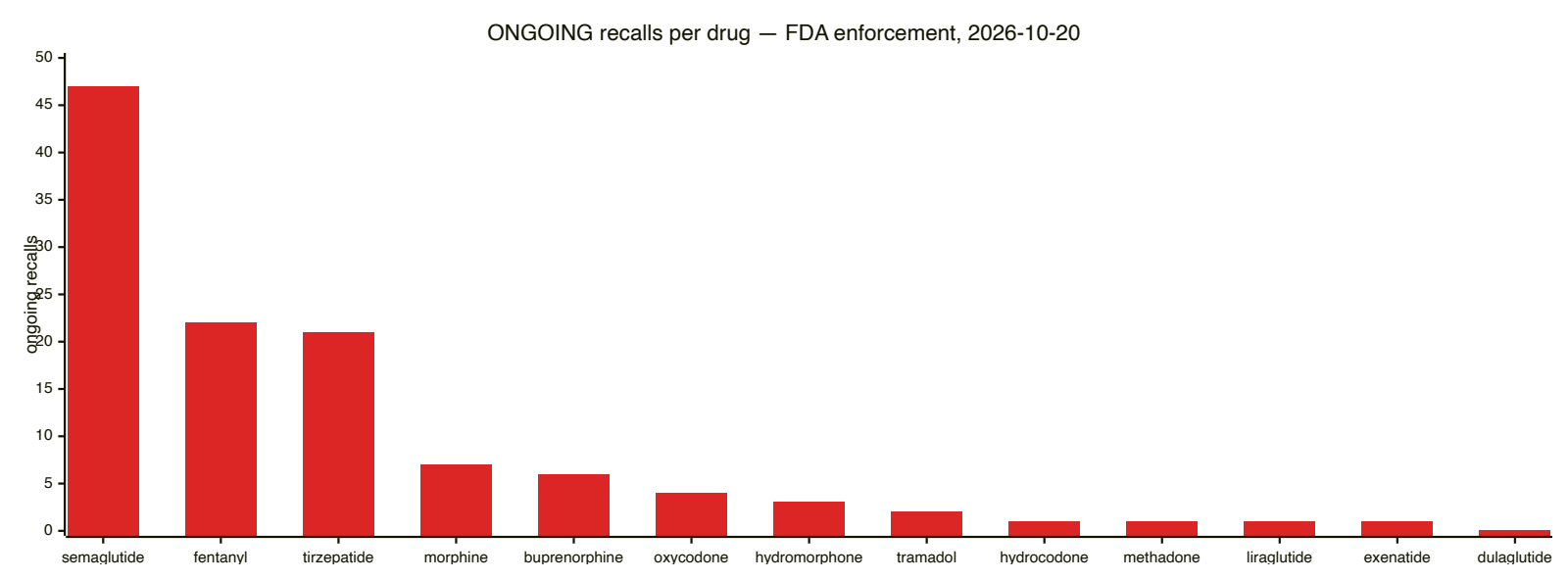


Example B — semaglutide + NAION: reported but not on the US label

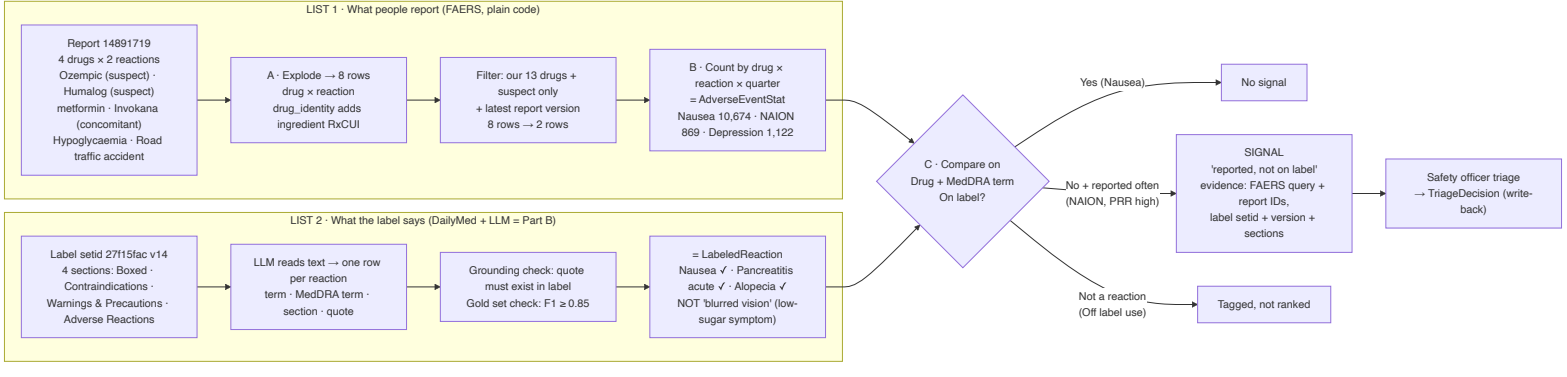


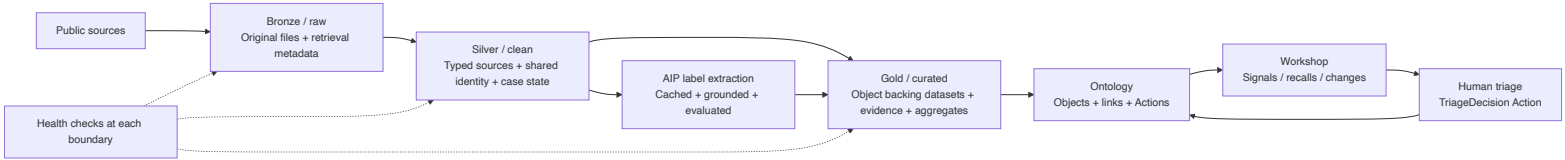

Join keys — what links the sources
FAERS
patient.drug[].openfda.rxcui
(product level)
openfda.spl_set_id ·
product_ndc
medicinalproduct (free text)
spl_set_id = setid (direct
link to label)
reaction term vs LLM-
extracted label text
reaction term (MedDRA)
rxcui roll-up to IN
DailyMed label
setid · spl_version
openfda.rxcui
Recalls
NDC parsed from
product_description
openfda.rxcui (almost
always empty)
ClinicalTrials.gov
intervention name (text
only)
rxcui roll-up
NDC → RxNorm; else text
match
name → approximateTerm
RxNorm ingredient RxCUI
THE one drug key

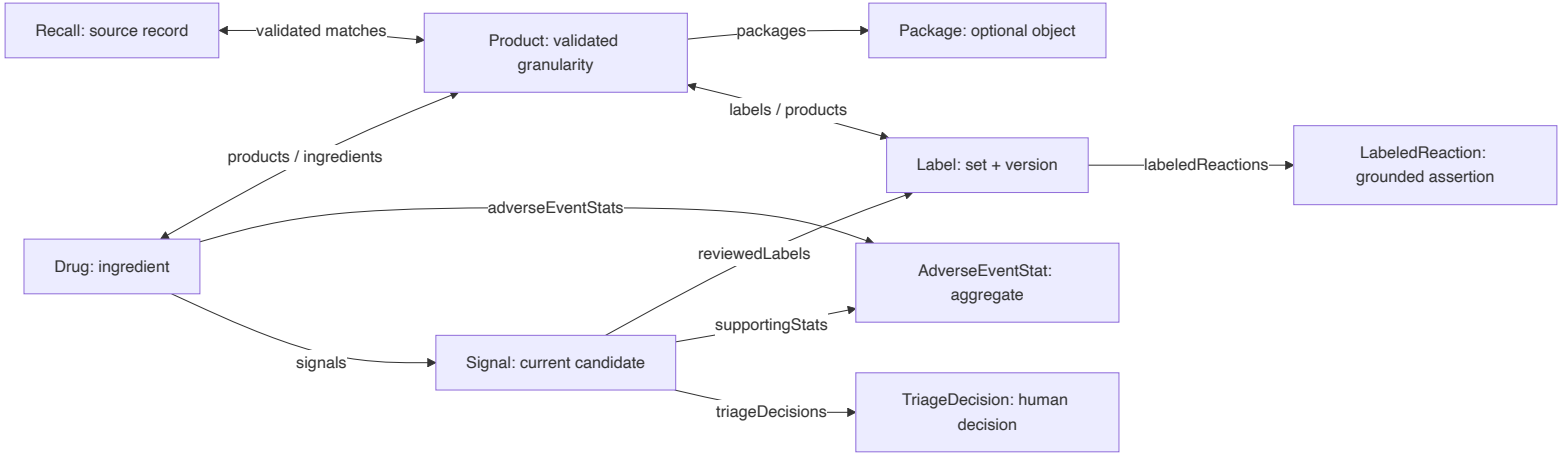

FAERS reports per drug (thousands) — openFDA, last updated 2026-07-30
Reports (k)
350
300
250
200
150
100
50
0
oxycodone
morphine
tirzepatide
buprenorphine
fentanyl
hydrocodone
hydromorphone
dulaglutide
semaglutide
exenatide
liraglutide
methadone
tramadol


ONGOING recalls per drug — FDA enforcement, 2026-10-20
ongoing recalls
50
45
40
35
30
25
20
15
10
5
0
semaglutide
fentanyl
tirzepatide
morphine
buprenorphine
oxycodone
hydromorphone
tramadol
hydrocodone
methadone
liraglutide
exenatide
dulaglutide

One FAERS report → List 1 · One label → List 2 (Part B) · Compare → Signal

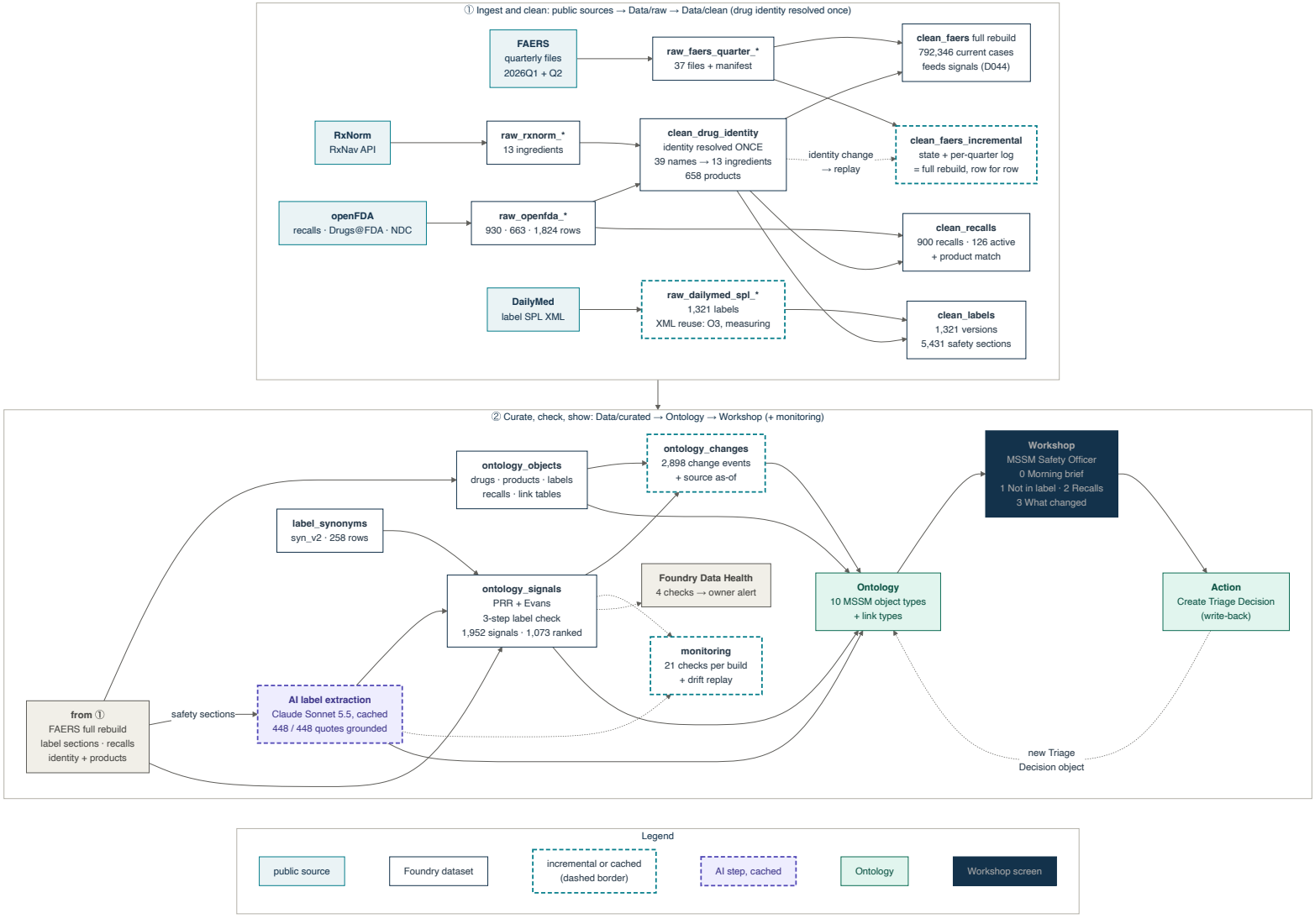



Public sources
Bronze / raw
Original files + retrieval metadata
Silver / clean
Typed sources + shared identity + case state
AIP label extraction
Cached + grounded + evaluated
Gold / curated
Object backing datasets + evidence + aggregates
Ontology
Objects + links + Actions
Workshop
Signals / recalls / changes
Human triage
TriageDecision Action
Health checks at each boundary

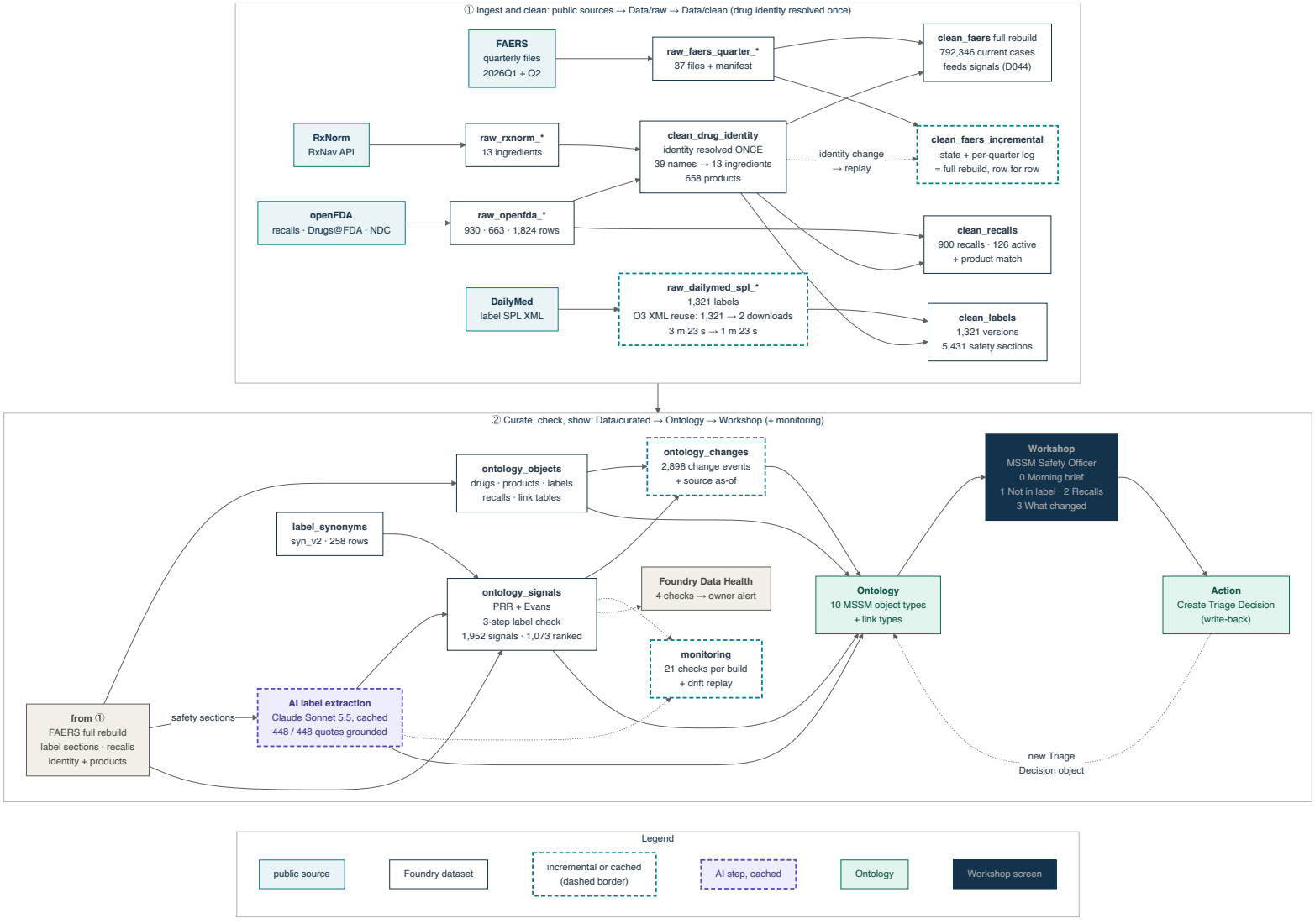

Recall: source record
validated matches
Product: validated granularity
packages
Package: optional object
labels / products
products / ingredients
Label: set + version
labeledReactions
LabeledReaction: grounded assertion
adverseEventStats
Drug: ingredient
reviewedLabels
AdverseEventStat: aggregate
supportingStats
signals
Signal: current candidate
triageDecisions
TriageDecision: human decision


Every capture: identity guard + sha256 + retrieval time
S1 RxNorm brand to ingredient RxCUI + reverse check
ingredient
S2 NDC Directory package, labeler, application no.
same application no.
S3 Drugs at FDA sponsor, products, strengths
spl_set_id from S2
S4 DailyMed label classify one phrase in one passage
is the change in this label?
S5 SrLC latest safety-label change
top terms checked against S4 label
S6 FAERS counts brand vs substance reports
does the recall name the S2 package?
S7 FDA enforcement latest recall
arm names the S1 ingredient
S8 ClinicalTrials.gov one arm: serious AEs affected / at risk
answers.jsonl 30 traces x 6 steps review pending
verify_trace.py re-derives every answer from raw bytes

Data flow as built (pilot, 13 drugs, 2026-11-05): public sources → raw → clean → curated → Ontology → Workshop



Data flow as built (pilot, 13 drugs, 2026-11-05): public sources → raw → clean → curated → Ontology → Workshop



Data flow as built v3 (pilot, 13 drugs, 2026-11-06): public sources → raw → clean → curated → Ontology → Workshop

① Ingest and clean: public sources → Data/raw → Data/clean (drug identity resolved once)

FAERS
quarterly files
2026Q1 + Q2

raw_faers_quarter_*
37 files + manifest

clean_faers full rebuild
792,346 current cases
feeds signals (D044)

clean_faers_incremental
state + per-quarter log
= full rebuild, row for row

RxNorm
RxNav API

raw_rxnorm_*
13 ingredients

clean_drug_identity
identity resolved ONCE
39 names → 13 ingredients
658 products

identity change
→ replay

openFDA
recalls · Drugs@FDA · NDC

raw_openfda_*
930 · 663 · 1,824 rows

clean_recalls
900 recalls · 126 active
+ product match

DailyMed
label SPL XML

raw_dailymed_spl_*
1,321 labels
O3 XML reuse: 1,321 → 2 downloads
3 m 23 s → 1 m 23 s

clean_labels
1,321 versions
5,431 safety sections

② Curate, check, show: Data/curated → Ontology → Workshop (+ monitoring)

ontology_objects
drugs · products · labels
recalls · link tables

ontology_changes
2,898 change events
+ source as-of

Workshop v0.10.0
MSSM Safety Officer
0 Morning brief
1 Not in label (+ AI quotes)
2 Recalls · 3 What changed

label_synonyms
syn_v2 · 258 rows

ontology_signals
PRR + Evans
3-step label check
1,952 signals · 1,073 ranked

Foundry Data Health
4 checks → owner alert

Ontology
10 MSSM object types
10 MSSM link types

Action
Create Triage Decision
(write-back)

monitoring
21 checks per build
+ drift replay

from ①
FAERS full rebuild
label sections · recalls
identity + products

safety sections

AI label extraction
Claude Sonnet 5.5, cached
5 labels: 2 oxycodone + 3 semaglutide
745 / 745 quotes grounded

new Triage
Decision object

Legend

public source

Foundry dataset

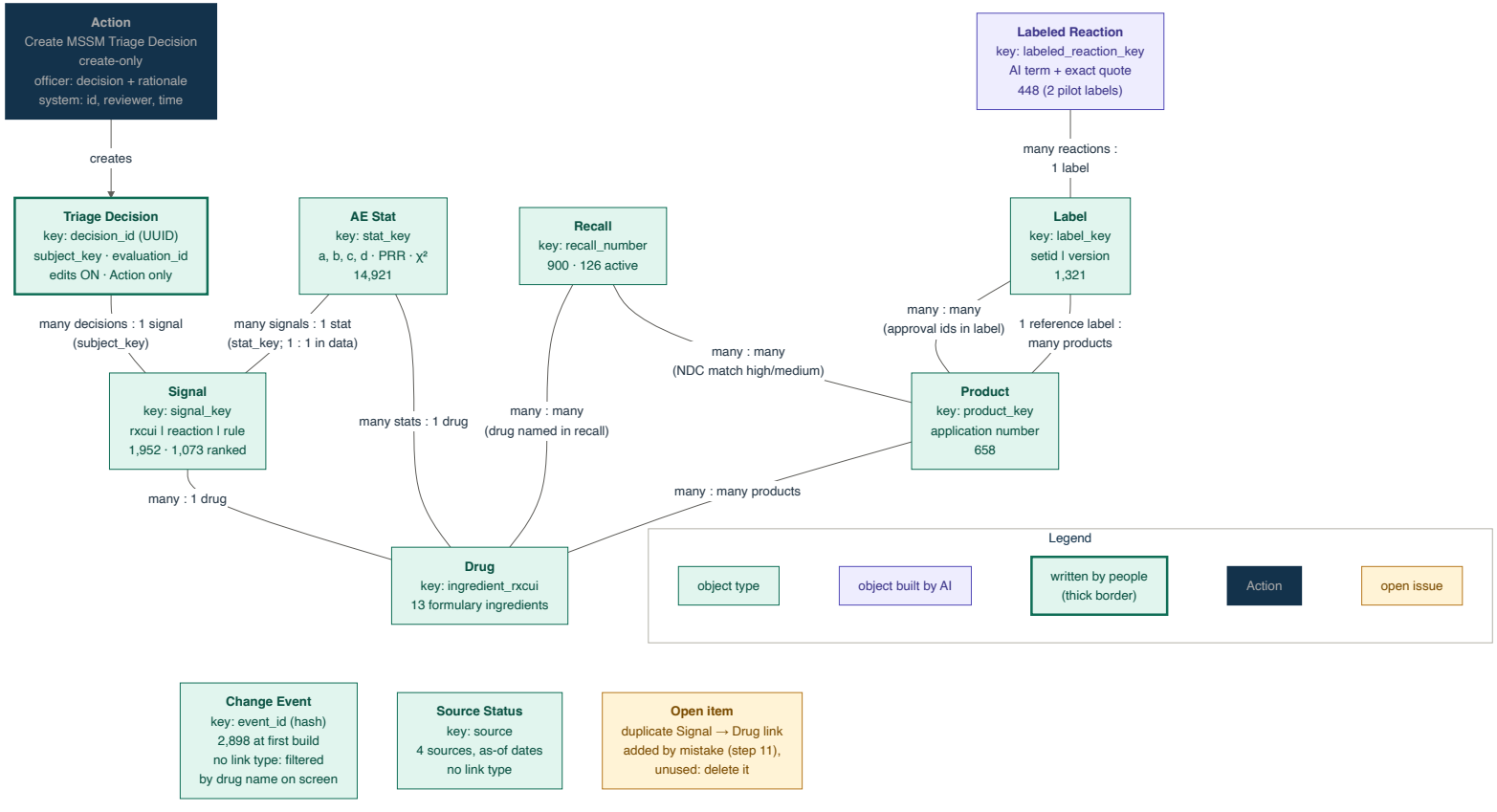
incremental or cached
(dashed border)

AI step, cached

Ontology

Workshop screen

Ontology as built: 10 MSSM object types, link types (cardinality on each line) and the one Action (rows = backing dataset, 2026-11-05)



Ontology as built v2: 10 MSSM object types, 10 link types (cardinality on each line) and the one Action (rows = backing dataset, 2026-11-06)

Action
Create MSSM Triage Decision
create-only
officer: decision + rationale
system: id, reviewer, time

creates

Triage Decision
key: decision_id (UUID)
subject_key · evaluation_id
edits ON · Action only

many decisions : 1 signal
(subject_key)

Signal
key: signal_key
rxcui | reaction | rule
1,952 · 1,073 ranked

many : 1 drug

AE Stat
key: stat_key
a, b, c, d · PRR · χ^2
14,921

many signals : 1 stat
(stat_key; 1 : 1 in data)

many stats : 1 drug

Recall
key: recall_number
900 · 126 active

many : many
(drug named in recall)

many : many
(NDC match high/medium)

Labeled Reaction
key: labeled_reaction_key
AI term + exact quote
745 (5 labels: 2 oxycodone
+ 3 semaglutide)

many reactions :
1 label

Label
key: label_key
setid | version
1,321

many : many
(approval ids in label)

1 reference label :
many products

Product
key: product_key
application number
658

many : many products

Drug
key: ingredient_rxcui
13 formulary ingredients

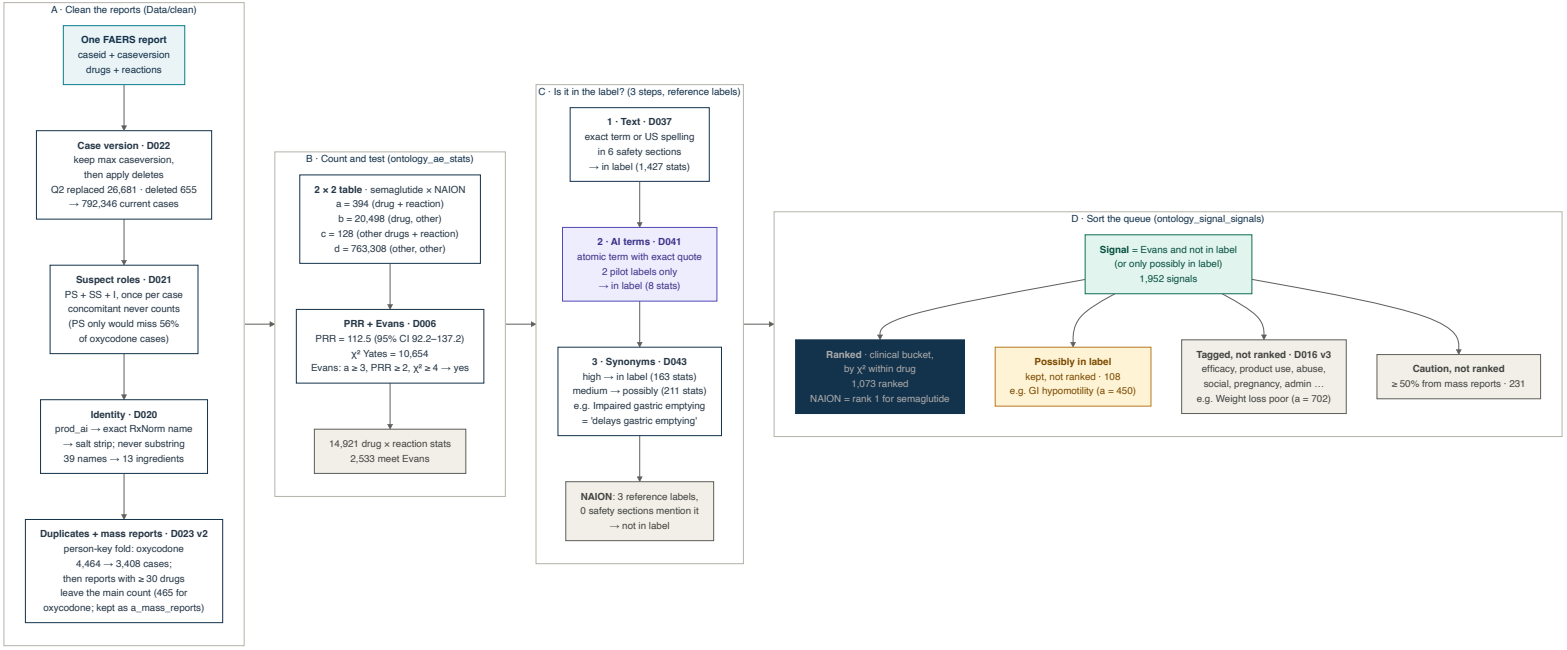
Legend

- object type
- object built by AI
- written by people (thick border)
- Action

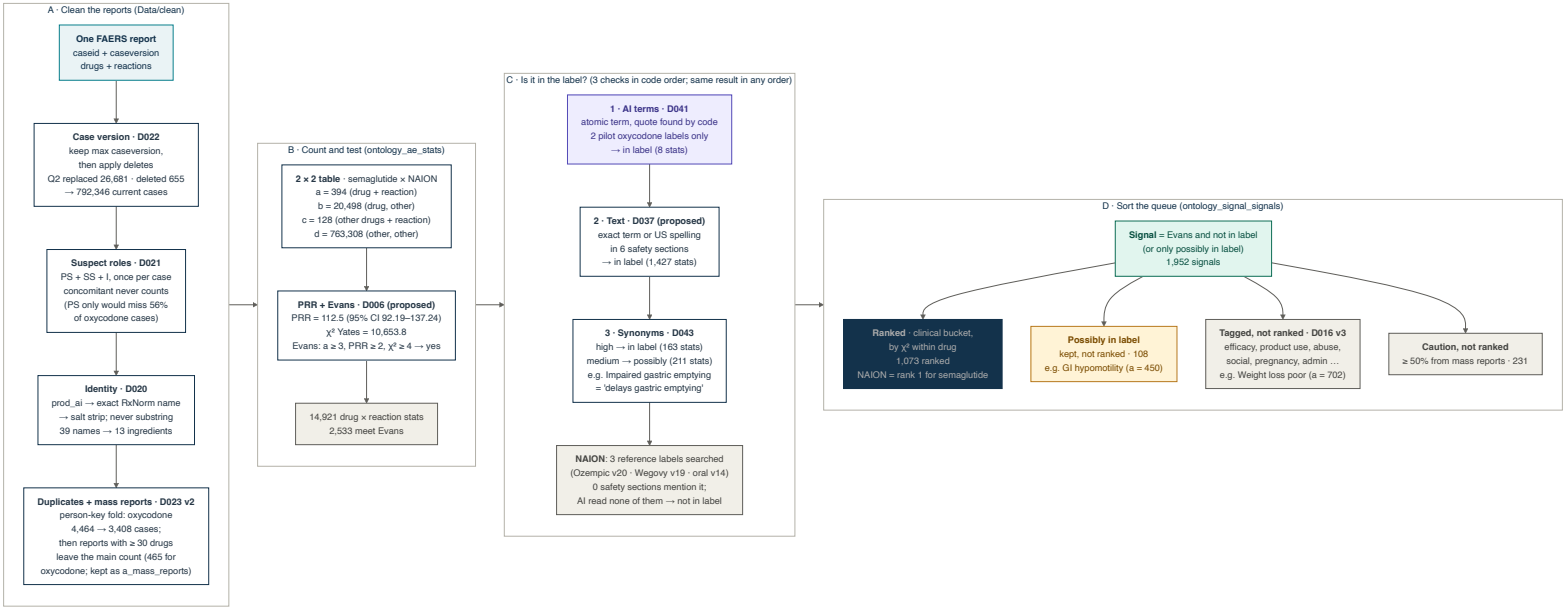
Change Event
key: event_id (hash)
2,898 at first build
no link type: filtered
by drug name on screen

Source Status
key: source
4 sources, as-of dates
no link type

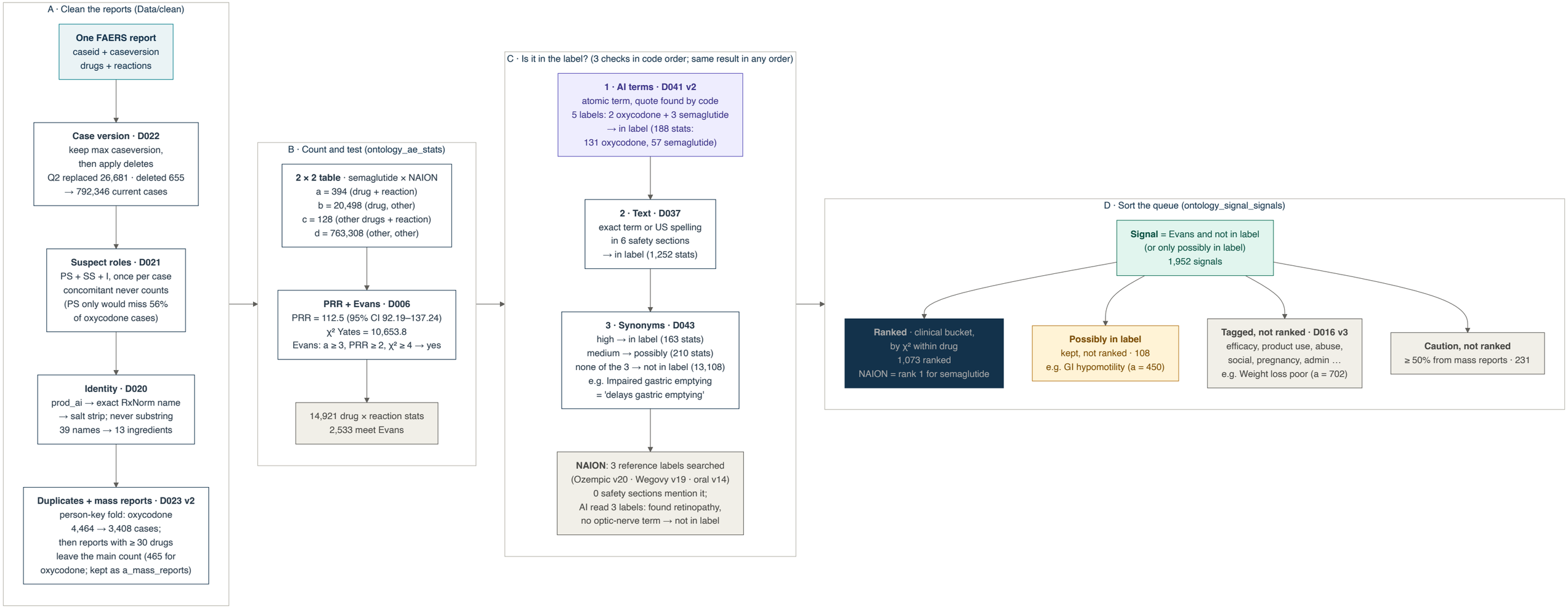
From one FAERS report to a ranked signal (rule prr_evans_v2, label check text+ai_v1+syn_v2) · example: semaglutide × NAION



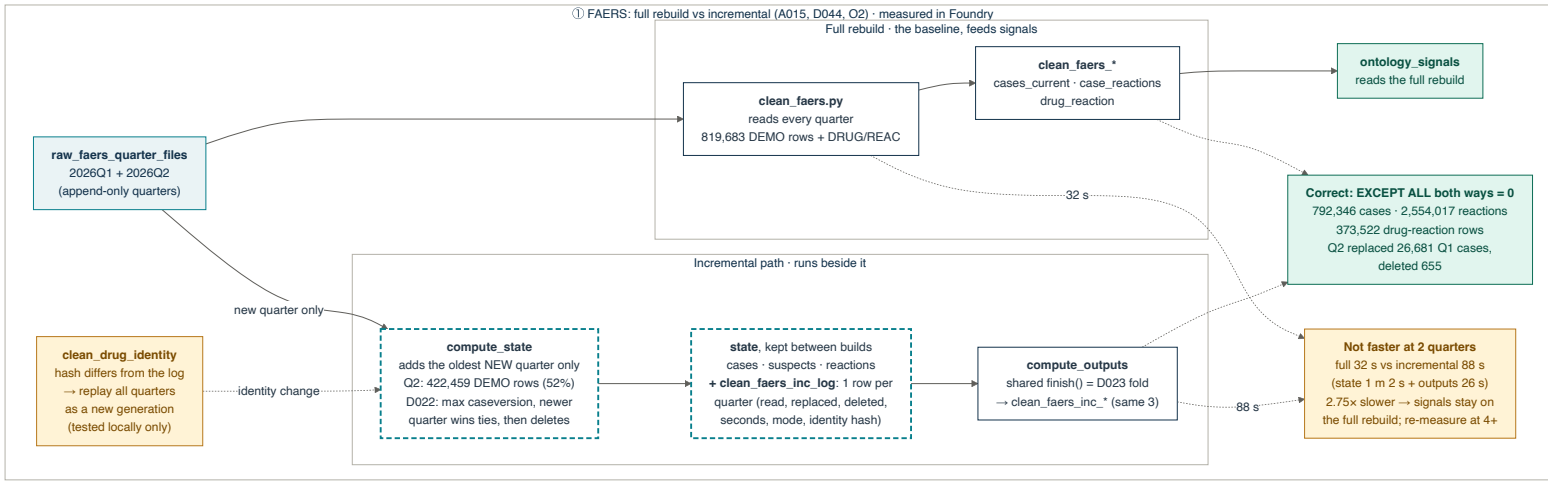
From one FAERS report to a ranked signal (rule prr_evans_v2, label check ai_v1 → text → syn_v2) · example: semaglutide × NAION

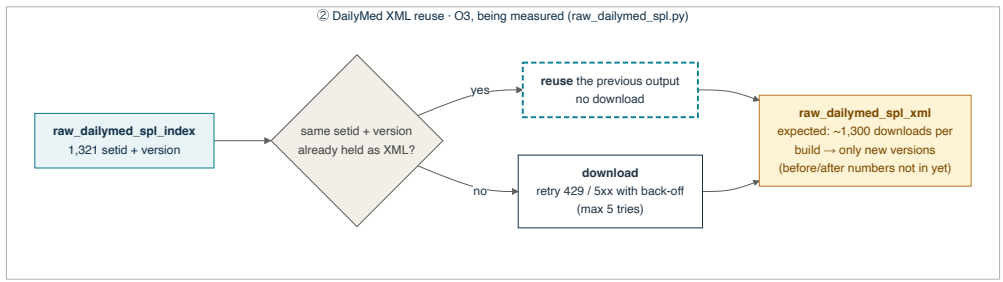


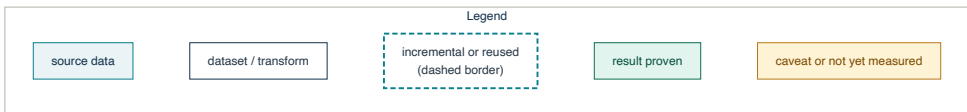
From one FAERS report to a ranked signal v3 (rule prr_evans_v2, label check ai_v1 → text → syn_v2; counts from Foundry, 2026-11-06) · example: semaglutide × NAION



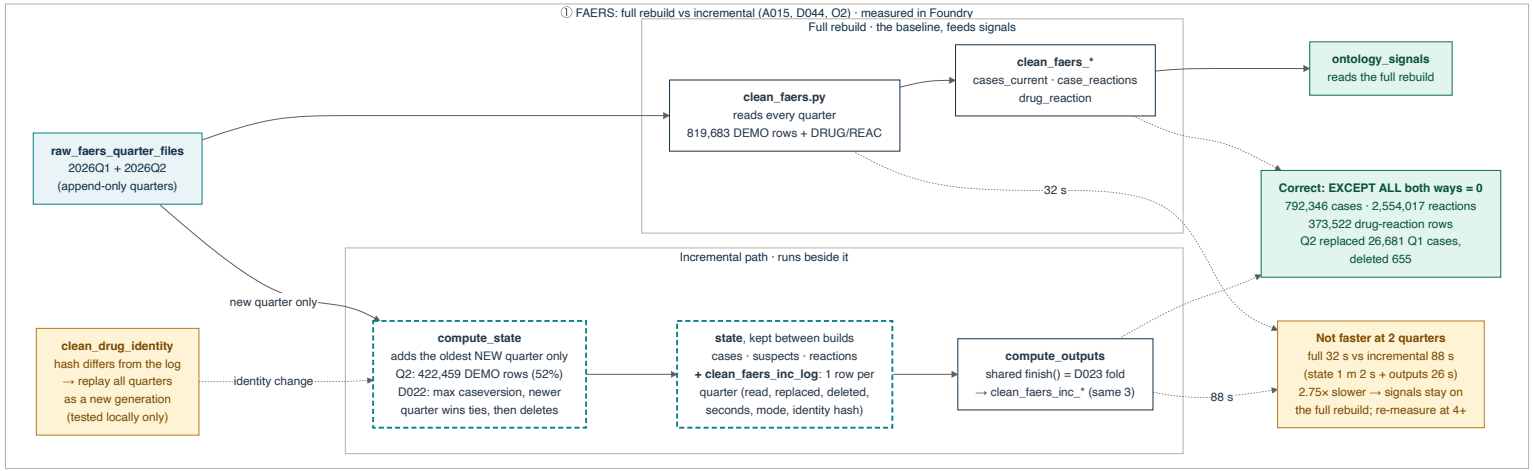
Incremental pieces: FAERS full vs incremental (A015, O2, measured) and DailyMed XML reuse (O3, being measured)

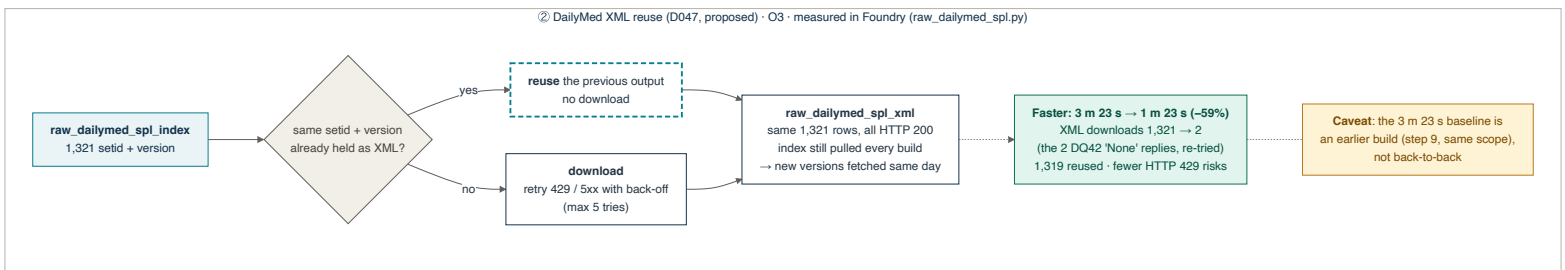






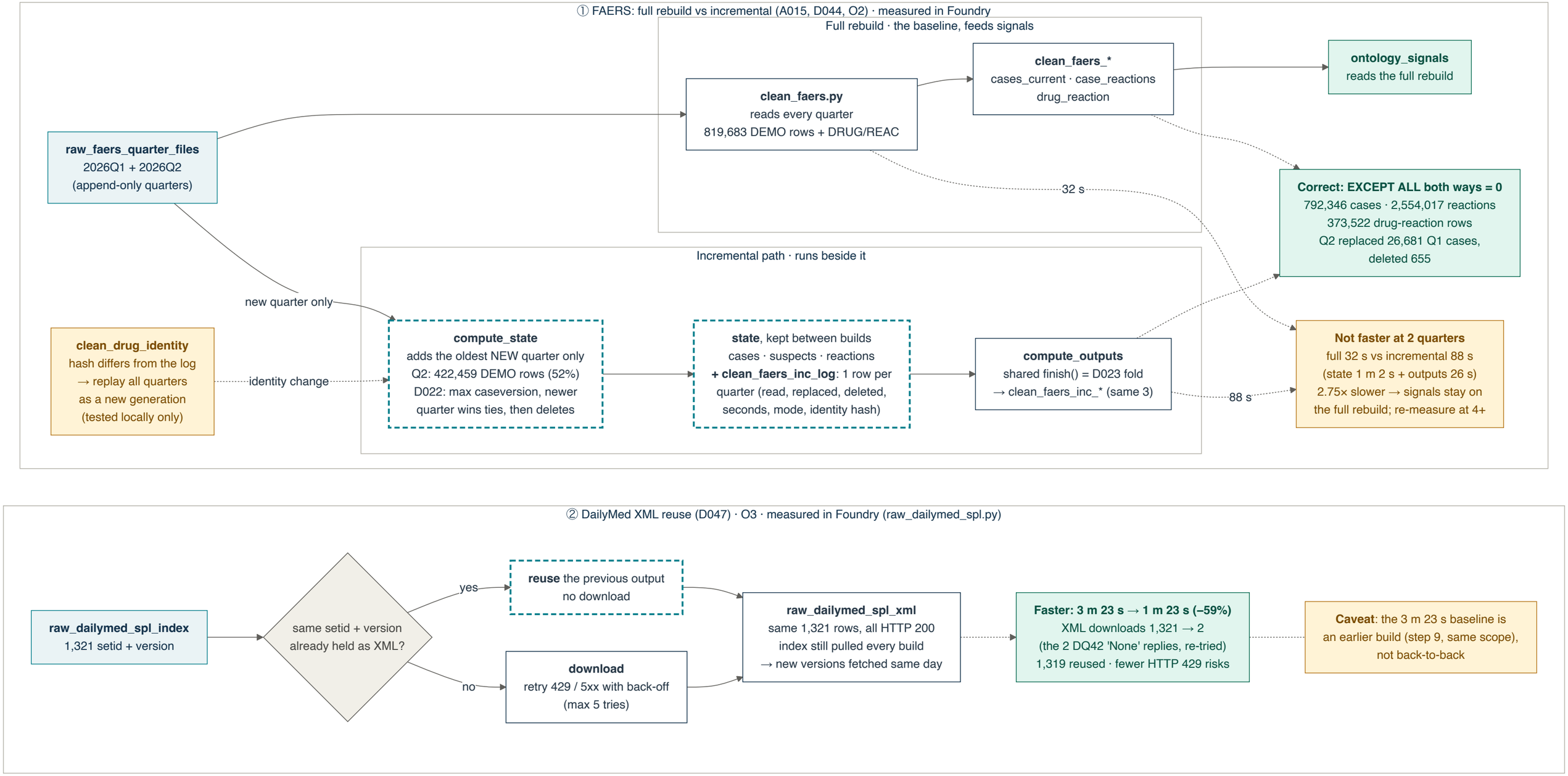
Incremental pieces: FAERS full vs incremental (A015, O2, measured: slower) and DailyMed XML reuse (O3, measured: faster)



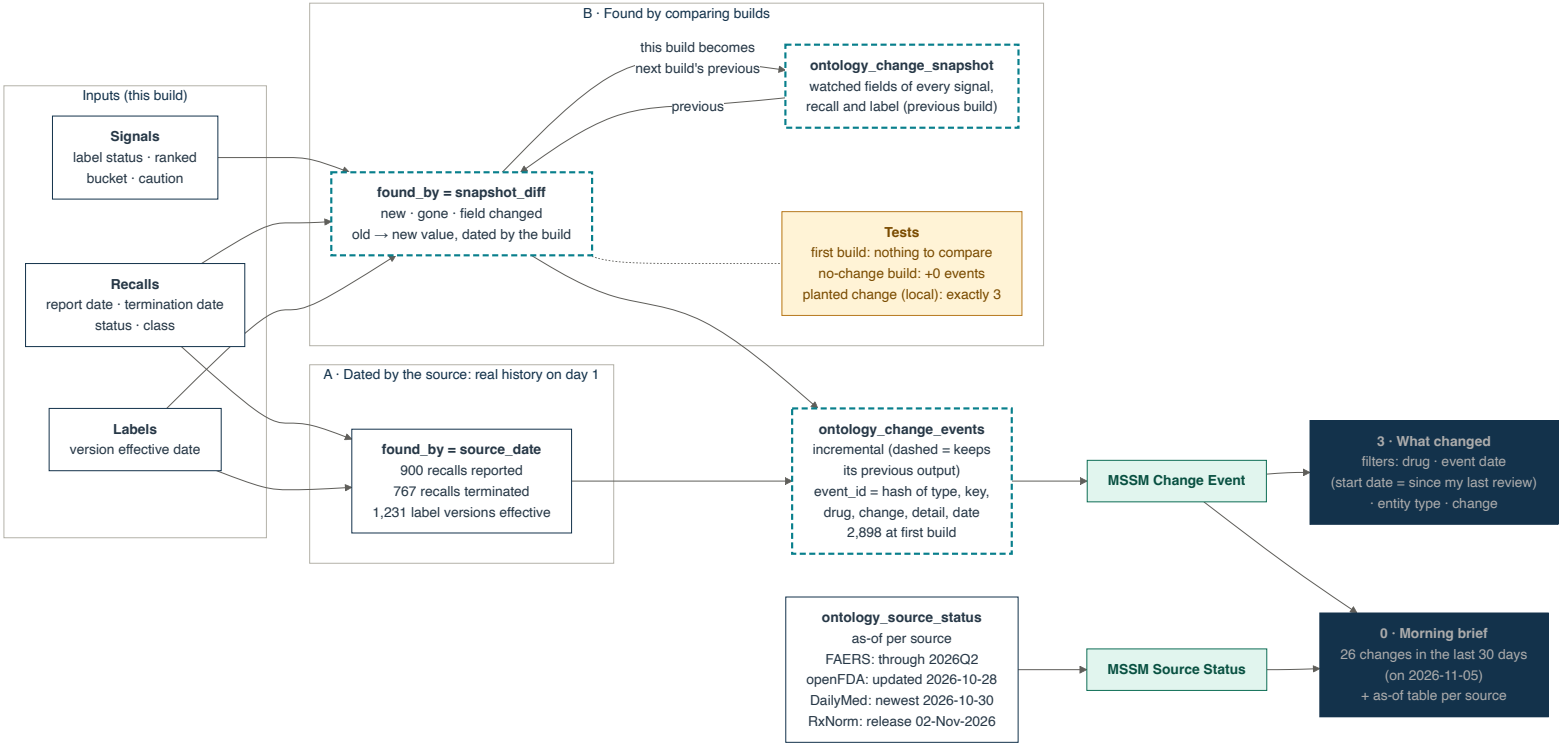


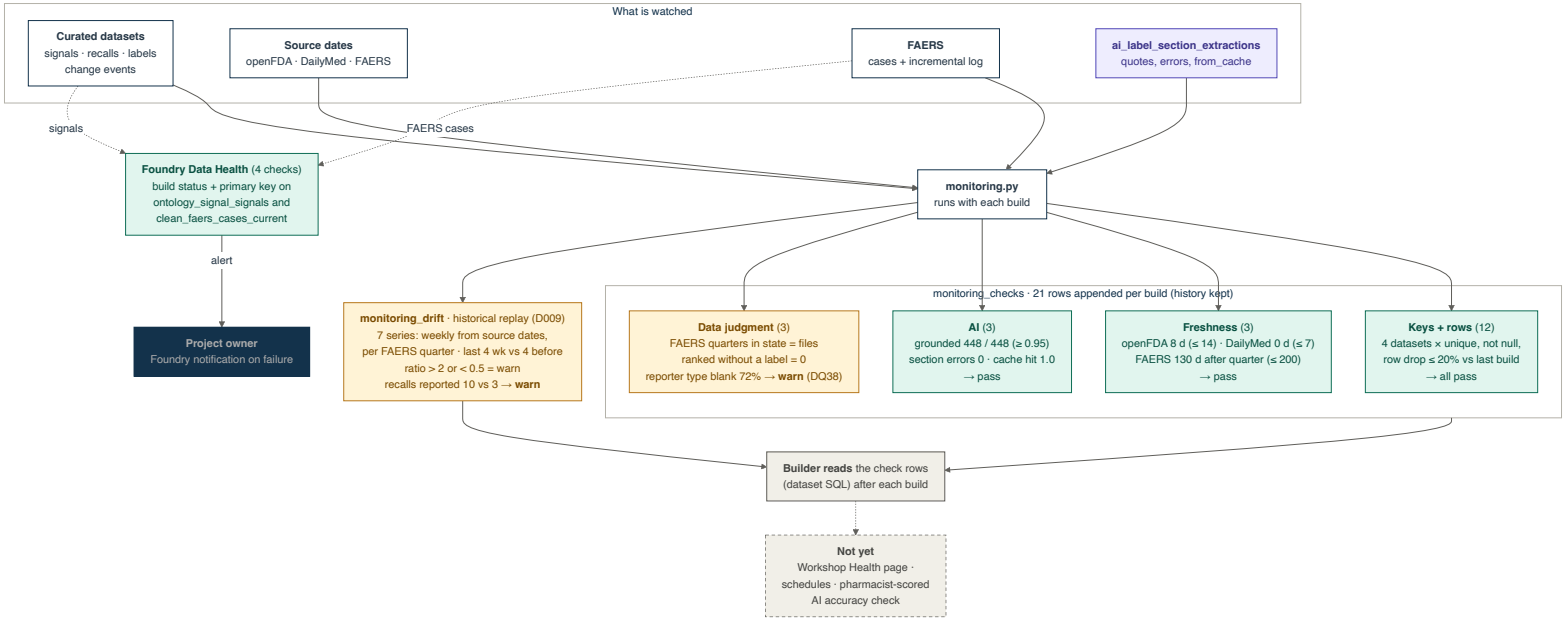


Incremental pieces v3: FAERS full vs incremental (A015, O2, measured: slower) and DailyMed XML reuse (O3, measured: faster)

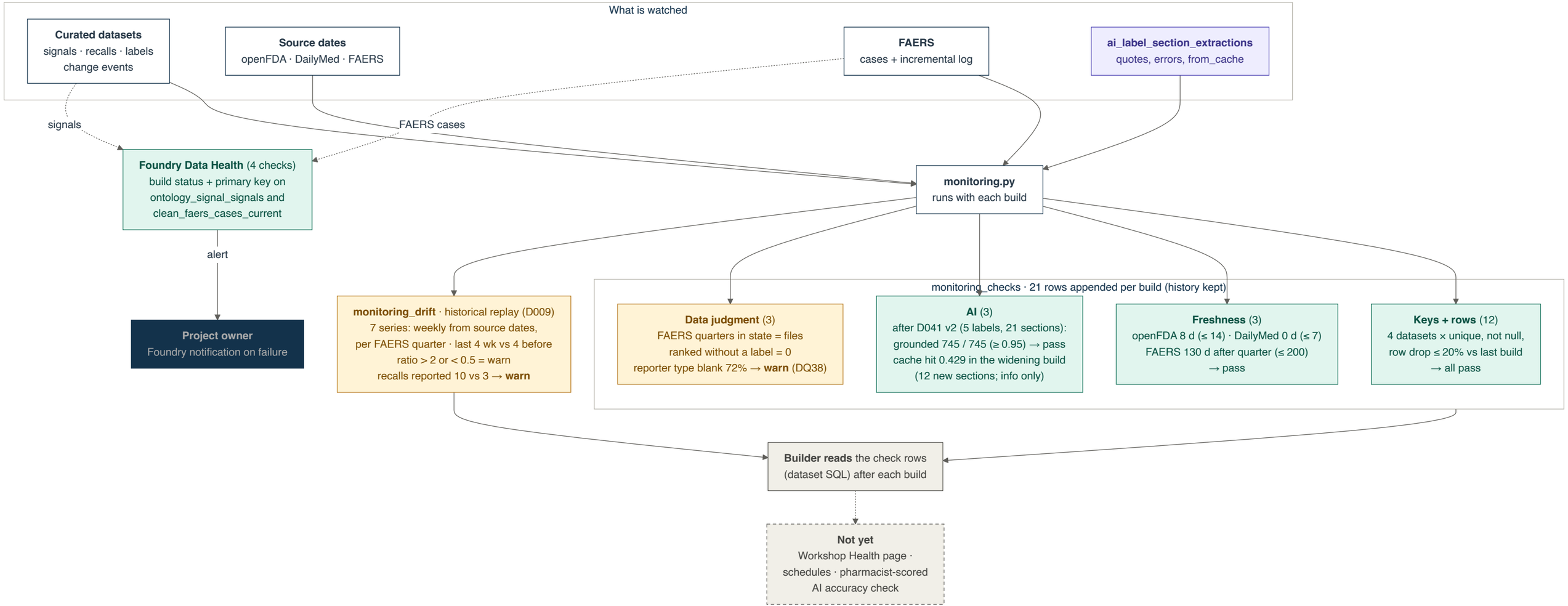


Change feed (D045, A016): two ways to know something changed → MSSM Change Event → page 3 and the Morning brief

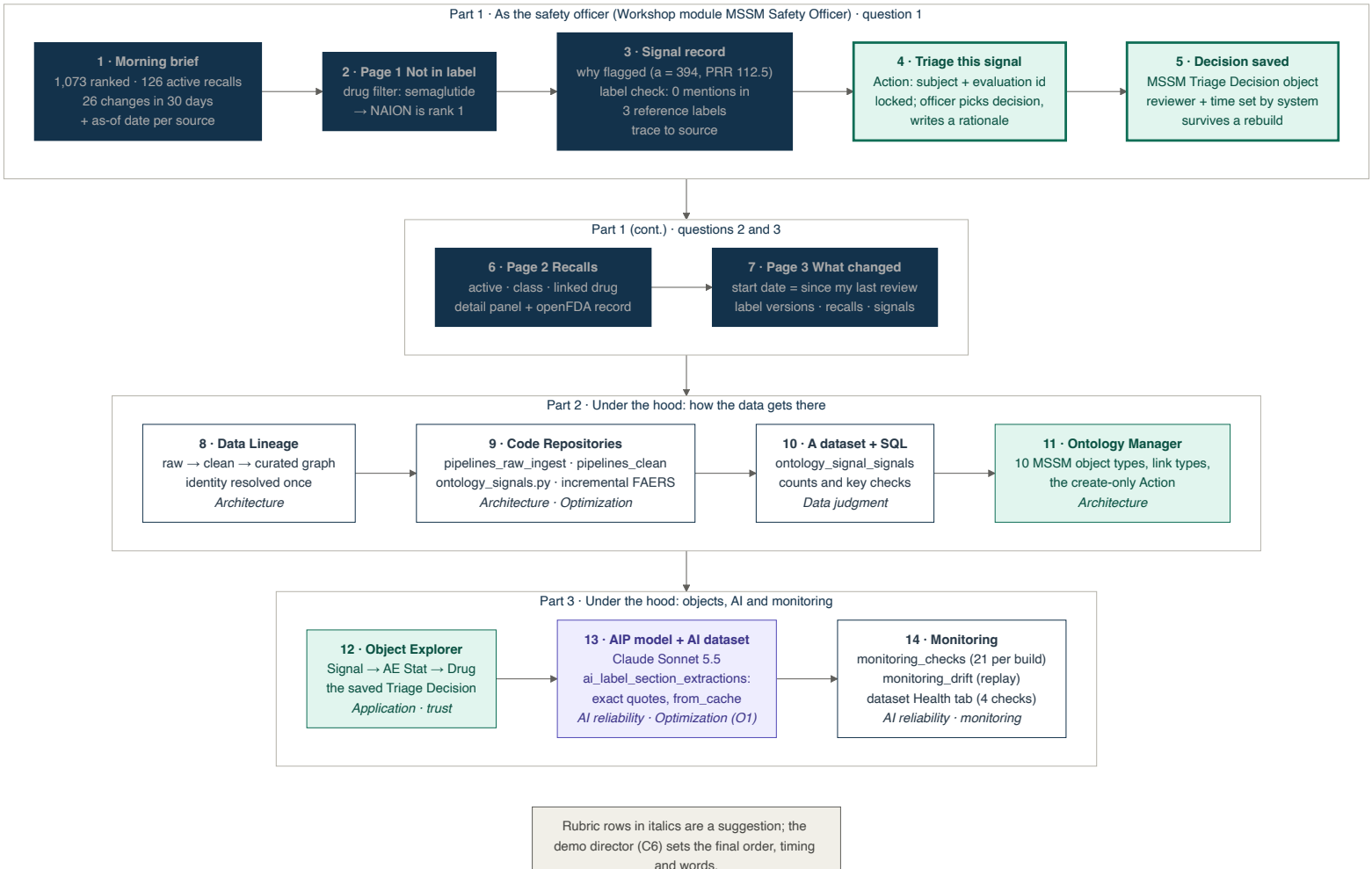


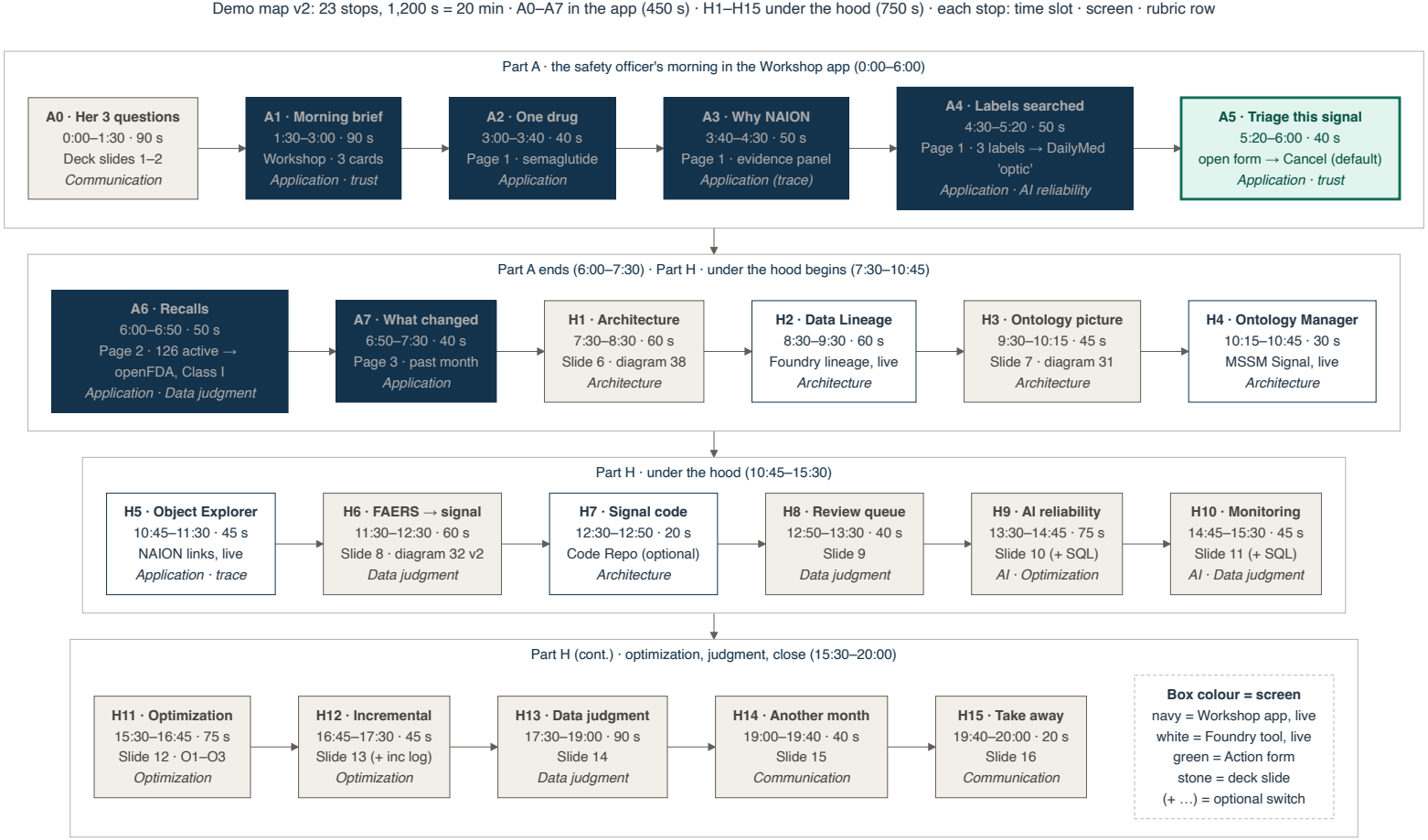

Monitoring (D046, A017): computed checks + drift replay + Foundry Data Health → who finds out (first Foundry run, 2026-11-05)
What is watched
Curated datasets
signals · recalls · labels
change events
Source dates
openFDA · DailyMed · FAERS
FAERS
cases + incremental log
ai_label_section_extractions
quotes, errors, from_cache
signals
FAERS cases
Foundry Data Health (4 checks)
build status + primary key on
ontology_signal_signals and
clean_faers_cases_current
monitoring.py
runs with each build
alert
Project owner
Foundry notification on failure
monitoring_drift · historical replay (D009)
7 series: weekly from source dates,
per FAERS quarter · last 4 wk vs 4 before
ratio > 2 or < 0.5 = warn
recalls reported 10 vs 3 → warn
monitoring_checks · 21 rows appended per build (history kept)
Data judgment (3)
FAERS quarters in state = files
ranked without a label = 0
reporter type blank 72% → warn (DQ38)
AI (3)
grounded 448 / 448 (≥ 0.95)
section errors 0 · cache hit 1.0
→ pass
Freshness (3)
openFDA 8 d (≤ 14) · DailyMed 0 d (≤ 7)
FAERS 130 d after quarter (≤ 200)
→ pass
Keys + rows (12)
4 datasets × unique, not null,
row drop ≤ 20% vs last build
→ all pass
Builder reads the check rows
(dataset SQL) after each build
Not yet
Workshop Health page ·
schedules · pharmacist-scored
AI accuracy check

Monitoring v2 (D046, A017): computed checks + drift replay + Foundry Data Health → who finds out (first Foundry run 2026-11-05; AI row updated after D041 v2, 2026-11-06)



Demo map: the officer's morning on screen, then the stops under the hood (rubric row each stop supports)




Demo map v2: 23 stops, 1,200 s = 20 min · A0–A7 in the app (450 s) · H1–H15 under the hood (750 s) · each stop: time slot · screen · rubric row
Part A · the safety officer's morning in the Workshop app (0:00–6:00)
A0 · Her 3 questions
0:00–1:30 · 90 s
Deck slides 1–2
Communication
A1 · Morning brief
1:30–3:00 · 90 s
Workshop · 3 cards
Application · trust
A2 · One drug
3:00–3:40 · 40 s
Page 1 · semaglutide
Application
A3 · Why NAION
3:40–4:30 · 50 s
Page 1 · evidence panel
Application (trace)
A4 · Labels searched
4:30–5:20 · 50 s
Page 1 · 3 labels → DailyMed
'optic'
Application · AI reliability
A5 · Triage this signal
5:20–6:00 · 40 s
open form → Cancel (default)
Application · trust
Part A ends (6:00–7:30) · Part H · under the hood begins (7:30–10:45)
A6 · Recalls
6:00–6:50 · 50 s
Page 2 · 126 active →
openFDA, Class I
Application · Data judgment
A7 · What changed
6:50–7:30 · 40 s
Page 3 · past month
Application
H1 · Architecture
7:30–8:30 · 60 s
Slide 6 · diagram 38
Architecture
H2 · Data Lineage
8:30–9:30 · 60 s
Foundry lineage, live
Architecture
H3 · Ontology picture
9:30–10:15 · 45 s
Slide 7 · diagram 31
Architecture
H4 · Ontology Manager
10:15–10:45 · 30 s
MSSM Signal, live
Architecture
Part H · under the hood (10:45–15:30)
H5 · Object Explorer
10:45–11:30 · 45 s
NAION links, live
Application · trace
H6 · FAERS → signal
11:30–12:30 · 60 s
Slide 8 · diagram 32 v2
Data judgment
H7 · Signal code
12:30–12:50 · 20 s
Code Repo (optional)
Architecture
H8 · Review queue
12:50–13:30 · 40 s
Slide 9
Data judgment
H9 · AI reliability
13:30–14:45 · 75 s
Slide 10 (+ SQL)
AI · Optimization
H10 · Monitoring
14:45–15:30 · 45 s
Slide 11 (+ SQL)
AI · Data judgment
Part H (cont.) · optimization, judgment, close (15:30–20:00)
H11 · Optimization
15:30–16:45 · 75 s
Slide 12 · O1–O3
Optimization
H12 · Incremental
16:45–17:30 · 45 s
Slide 13 (+ inc log)
Optimization
H13 · Data judgment
17:30–19:00 · 90 s
Slide 14
Data judgment
H14 · Another month
19:00–19:40 · 40 s
Slide 15
Communication
H15 · Take away
19:40–20:00 · 20 s
Slide 16
Communication
Box colour = screen
navy = Workshop app, live
white = Foundry tool, live
green = Action form
stone = deck slide
(+ …) = optional switch

Demo map v3: 23 stops, 1,200 s = 20 min · A0–A7 in the app (450 s) · H1–H15 under the hood (750 s) · each stop: time slot · screen · rubric row

Part A · the safety officer's morning in the Workshop app (0:00–6:00)

A0 · Her 3 questions
0:00–1:30 · 90 s
Deck slides 1–2
Communication

→

A1 · Morning brief
1:30–3:00 · 90 s
Workshop · 3 cards
Application · trust

→

A2 · One drug
3:00–3:40 · 40 s
Page 1 · semaglutide
Application

→

A3 · Why NAION
3:40–4:30 · 50 s
Page 1 · evidence panel
Application (trace)

→

A4 · Labels searched
4:30–5:20 · 50 s
Page 1 · 3 labels → DailyMed
'optic'
→ AI quotes: retinopathy, no optic nerve
Application · AI reliability

→

A5 · Triage this signal
5:20–6:00 · 40 s
open form → Cancel (default)
Application · trust

Part A ends (6:00–7:30) · Part H · under the hood begins (7:30–10:45)

A6 · Recalls
6:00–6:50 · 50 s
Page 2 · 126 active →
openFDA, Class I
Application · Data judgment

→

A7 · What changed
6:50–7:30 · 40 s
Page 3 · past month
Application

→

H1 · Architecture
7:30–8:30 · 60 s
Slide 6 · diagram 38
Architecture

→

H2 · Data Lineage
8:30–9:30 · 60 s
Foundry lineage, live
Architecture

→

H3 · Ontology picture
9:30–10:15 · 45 s
Slide 7 · diagram 31
Architecture

→

H4 · Ontology Manager
10:15–10:45 · 30 s
MSSM Signal, live
Architecture

Part H · under the hood (10:45–15:30)

H5 · Object Explorer
10:45–11:30 · 45 s
NAION links, live
Application · trace

→

H6 · FAERS → signal
11:30–12:30 · 60 s
Slide 8 · diagram 32 v3
Data judgment

→

H7 · Signal code
12:30–12:50 · 20 s
Code Repo (optional)
Architecture

→

H8 · Review queue
12:50–13:30 · 40 s
Slide 9
Data judgment

→

H9 · AI reliability
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AI · Optimization

→

H10 · Monitoring
14:45–15:30 · 45 s
Slide 11 (+ SQL)
AI · Data judgment

Part H (cont.) · optimization, judgment, close (15:30–20:00)

H11 · Optimization
15:30–16:45 · 75 s
Slide 12 · O1–O3
Optimization

→

H12 · Incremental
16:45–17:30 · 45 s
Slide 13 (+ inc log)
Optimization

→

H13 · Data judgment
17:30–19:00 · 90 s
Slide 14
Data judgment

→

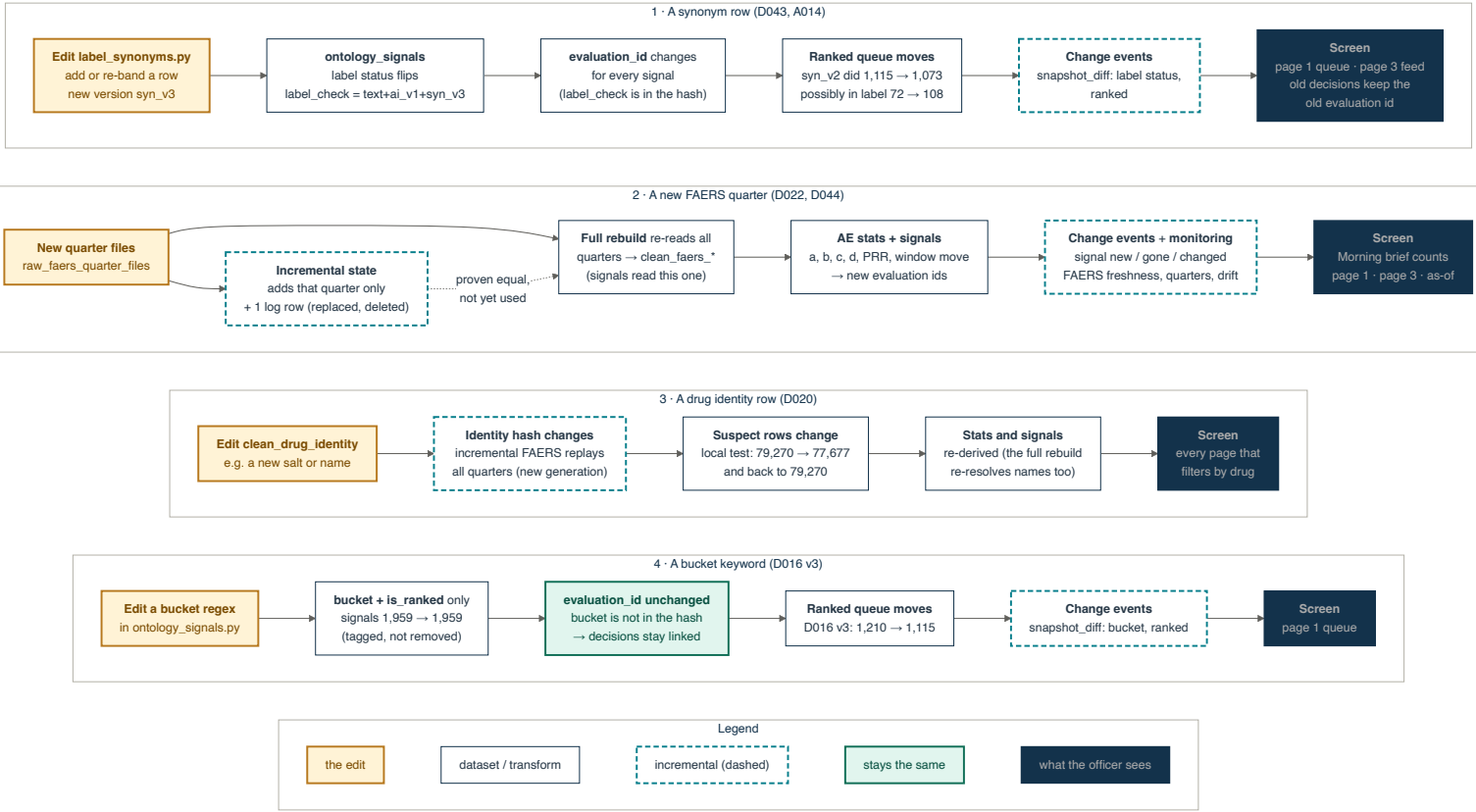
H14 · Another month
19:00–19:40 · 40 s
Slide 15
Communication

→

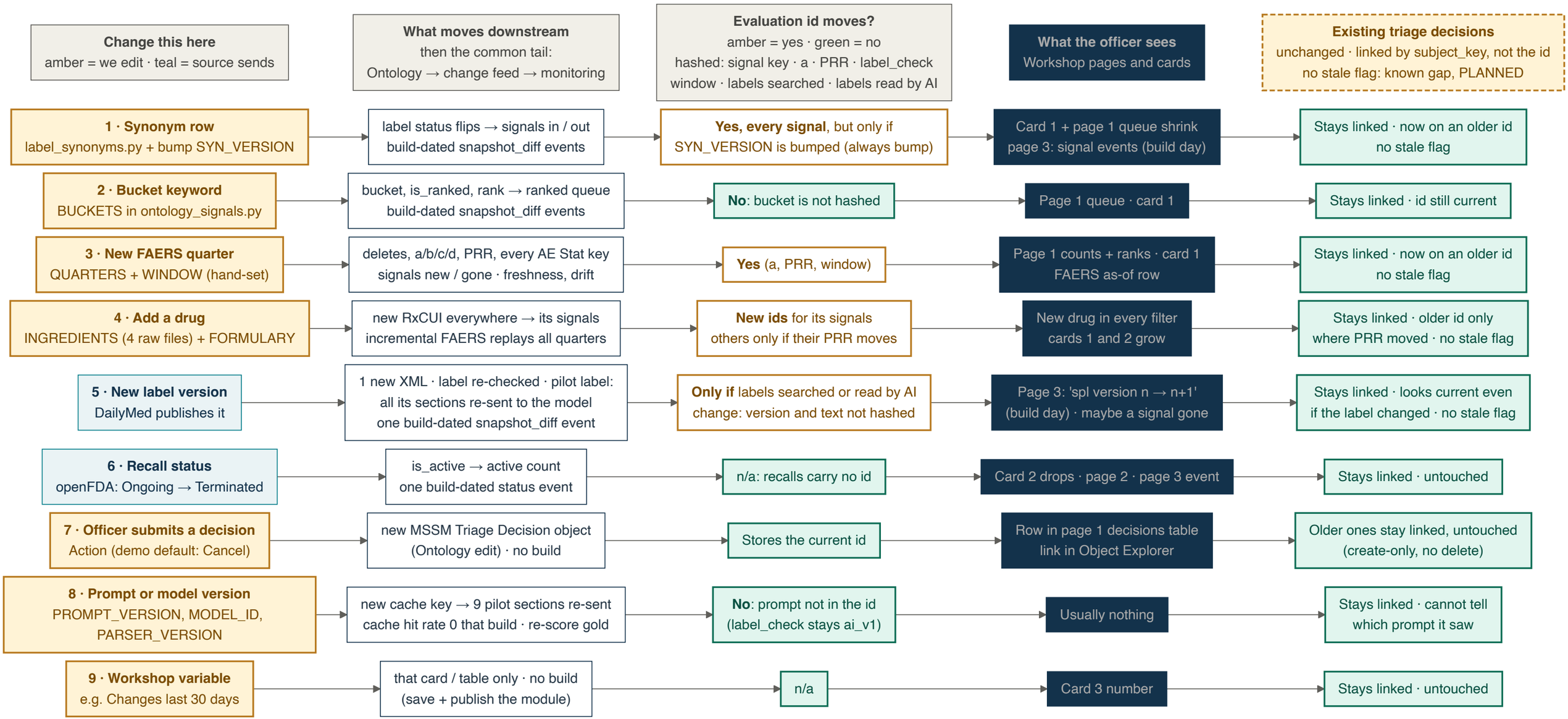
H15 · Take away
19:40–20:00 · 20 s
Slide 16
Communication

Box colour = screen
navy = Workshop app, live
white = Foundry tool, live
green = Action form
stone = deck slide
(+ …) = optional switch

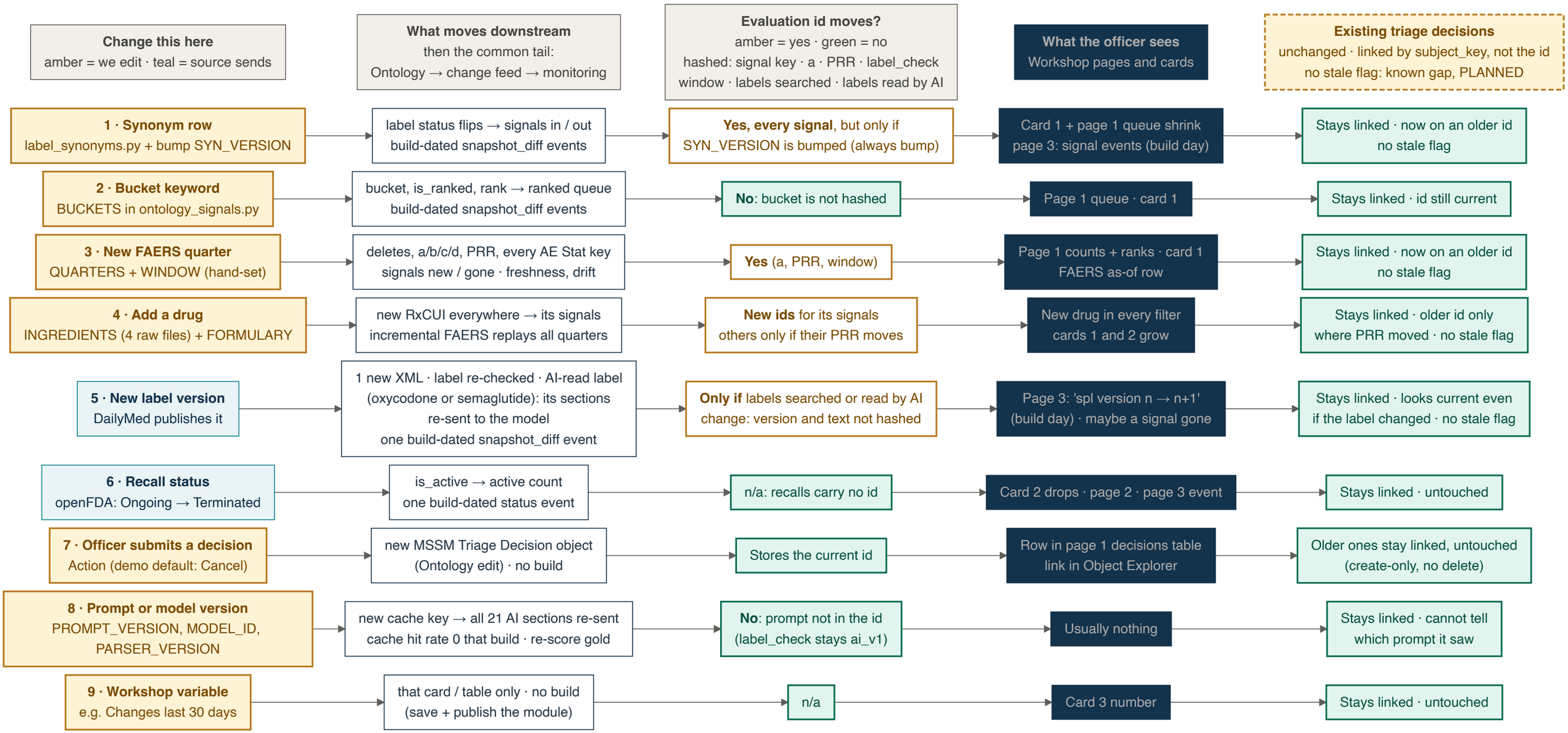
Ripple map: change this here → this changes there (four changes we made or can make, traced to the screen)

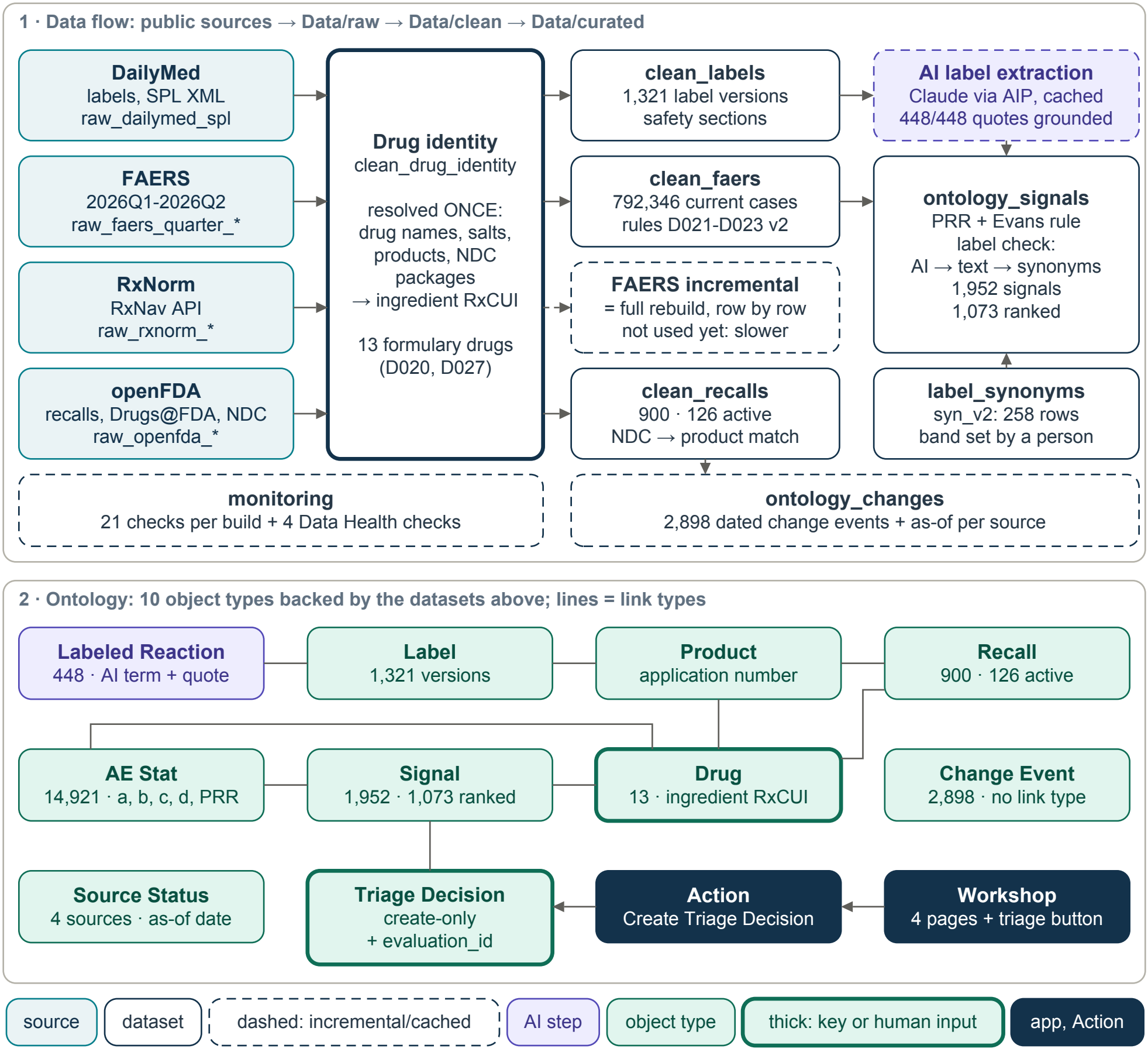


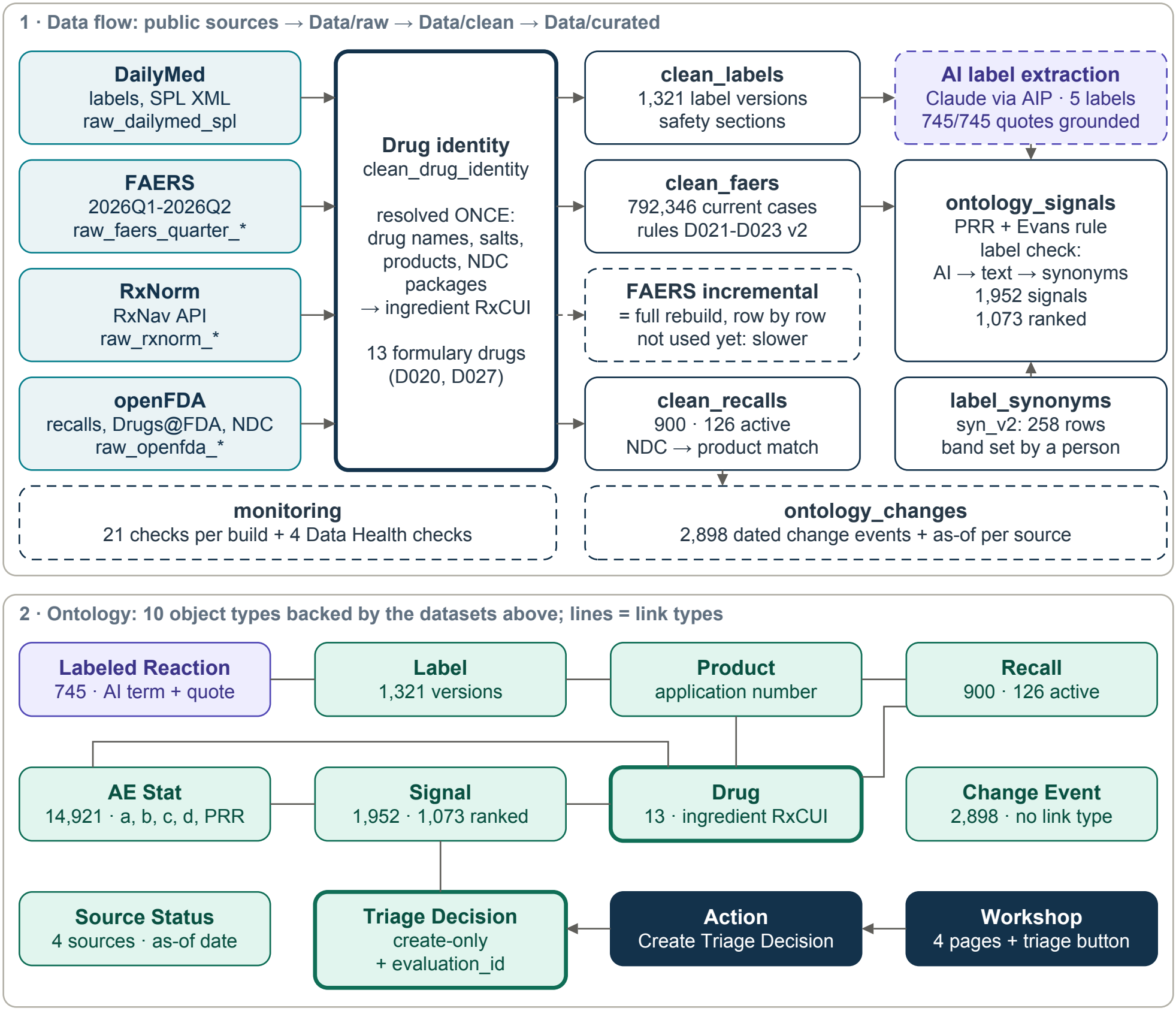
Ripple map v2: change this here → this changes there (9 changes, checked against the code) · existing triage decisions in every row



Ripple map v3: change this here → this changes there (9 changes, checked against the code) · existing triage decisions in every row




1 · Data flow: public sources → Data/raw → Data/clean → Data/curated
DailyMed
labels, SPL XML
raw_dailymed_spl
FAERS
2026Q1-2026Q2
raw_faers_quarter_*
RxNorm
RxNav API
raw_rxnorm_*
openFDA
recalls, Drugs@FDA, NDC
raw_openfda_*
Drug identity
clean_drug_identity
resolved ONCE:
drug names, salts,
products, NDC
packages
→ ingredient RxCUI
13 formulary drugs
(D020, D027)
clean_labels
1,321 label versions
safety sections
clean_faers
792,346 current cases
rules D021-D023 v2
FAERS incremental
= full rebuild, row by row
not used yet: slower
clean_recalls
900 · 126 active
NDC → product match
AI label extraction
Claude via AIP, cached
448/448 quotes grounded
ontology_signals
PRR + Evans rule
label check:
AI → text → synonyms
1,952 signals
1,073 ranked
label_synonyms
syn_v2: 258 rows
band set by a person
monitoring
21 checks per build + 4 Data Health checks
ontology_changes
2,898 dated change events + as-of per source
2 · Ontology: 10 object types backed by the datasets above; lines = link types
Labeled Reaction
448 · AI term + quote
Label
1,321 versions
Product
application number
Recall
900 · 126 active
AE Stat
14,921 · a, b, c, d, PRR
Signal
1,952 · 1,073 ranked
Drug
13 · ingredient RxCUI
Change Event
2,898 · no link type
Source Status
4 sources · as-of date
Triage Decision
create-only
+ evaluation_id
Action
Create Triage Decision
Workshop
4 pages + triage button
source
dataset
dashed: incremental/cached
AI step
object type
thick: key or human input
app, Action



source | dataset | dashed: incremental/cached | AI step | object type | thick: key or human input | app, Action